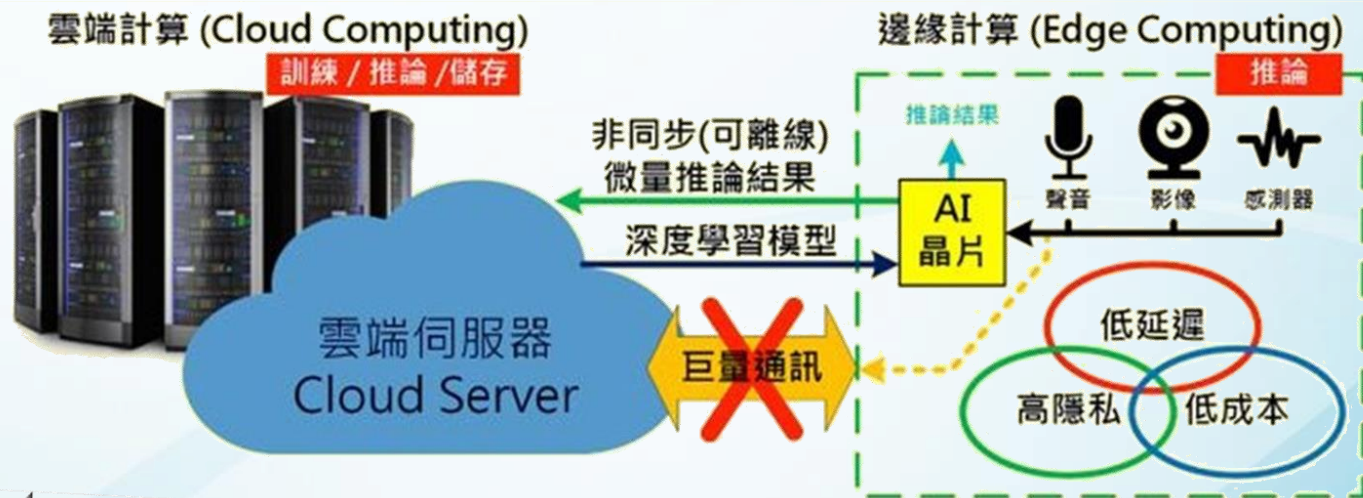
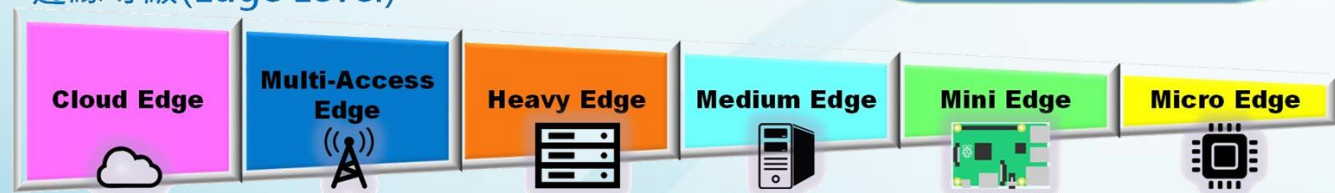


邊緣人工智慧實務 (EE5354701)



Edge AI

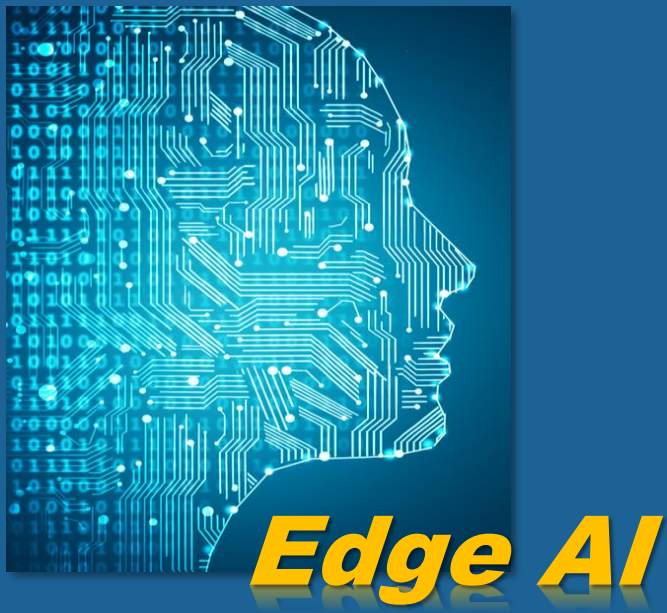
邊緣等級 (Edge Level)



8 TinyML 案例實作 — 聲音辨識

電機工程系 人工智慧應用產碩專班 許哲豪 兼任助理教授

簡報大綱

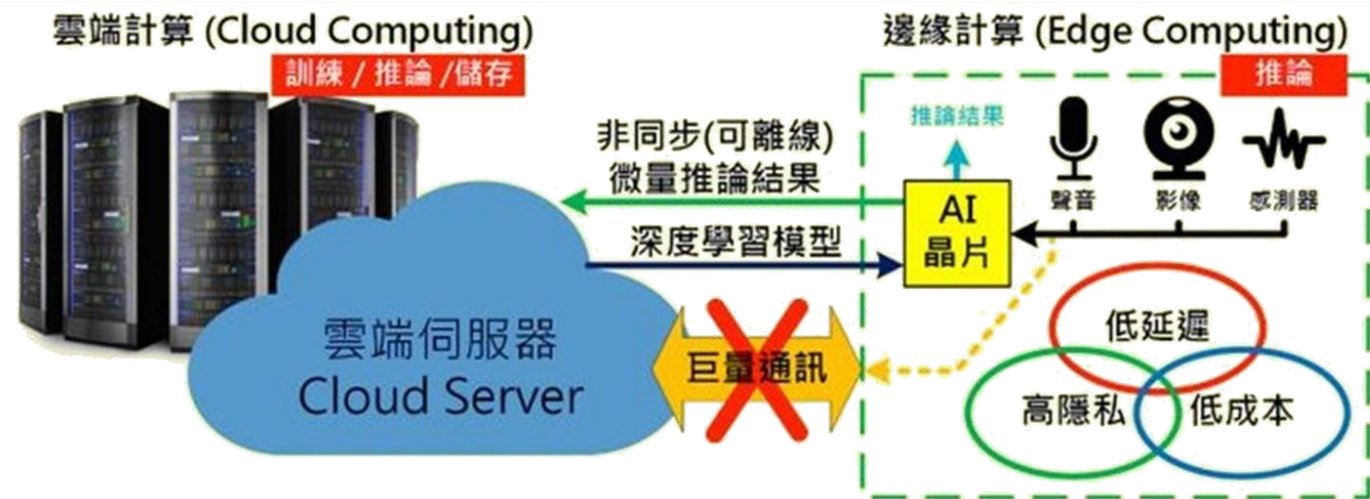


- 8.1. 聲音感測器原理
- 8.2. 喚醒詞偵測
- 8.3. 環境音辨識

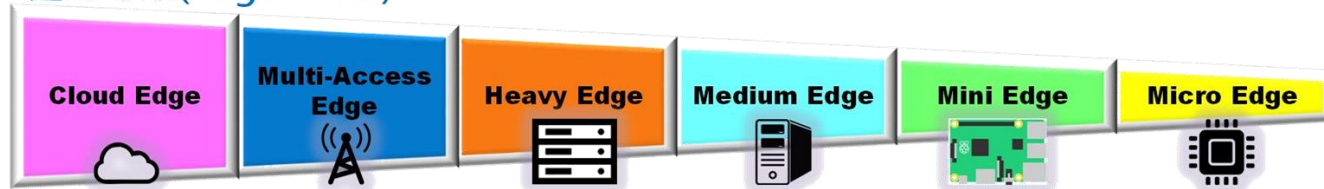
https://hackmd.io/@OmniXRI-Jack/NTUST_EdgeAI_2026



Edge AI

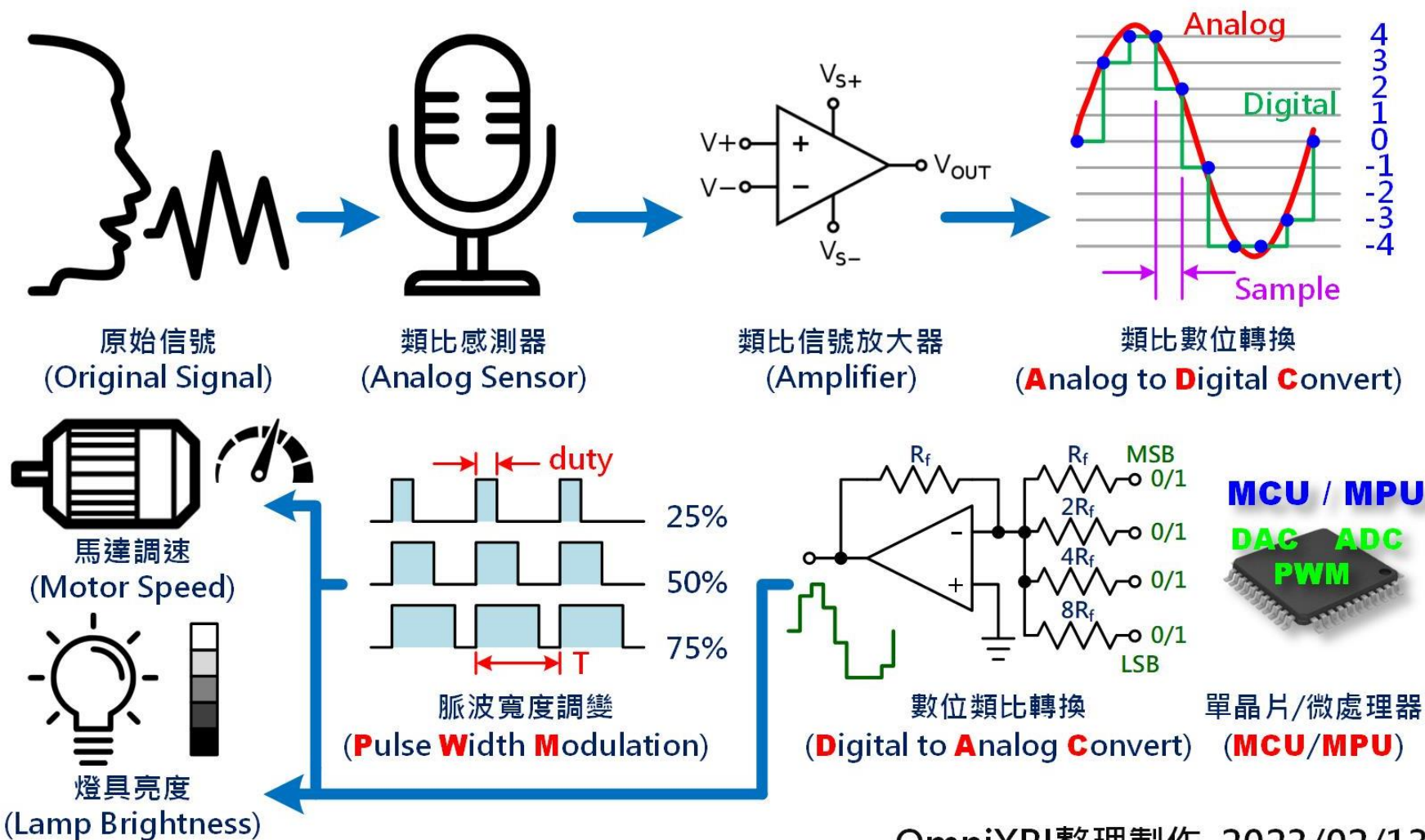


邊緣等級(Edge Level)



8.1. 聲音感測器原理

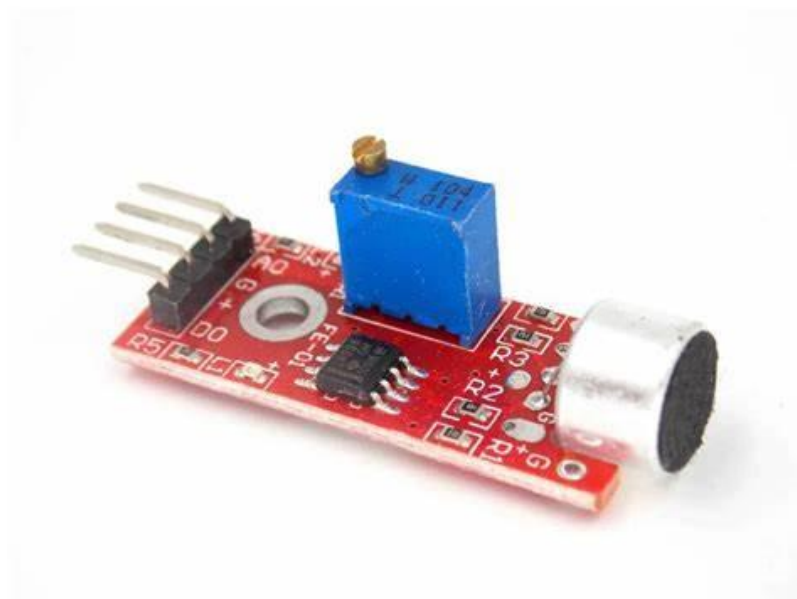
感測器類比/數位信號互換



資料來源：<https://omnixri.blogspot.com/2023/02/vmaker-edge-ai-02-ai.html>

OmniXRI整理製作, 2023/02/13

常見錄音裝置



電容式類比麥克風
(類比電壓輸出)

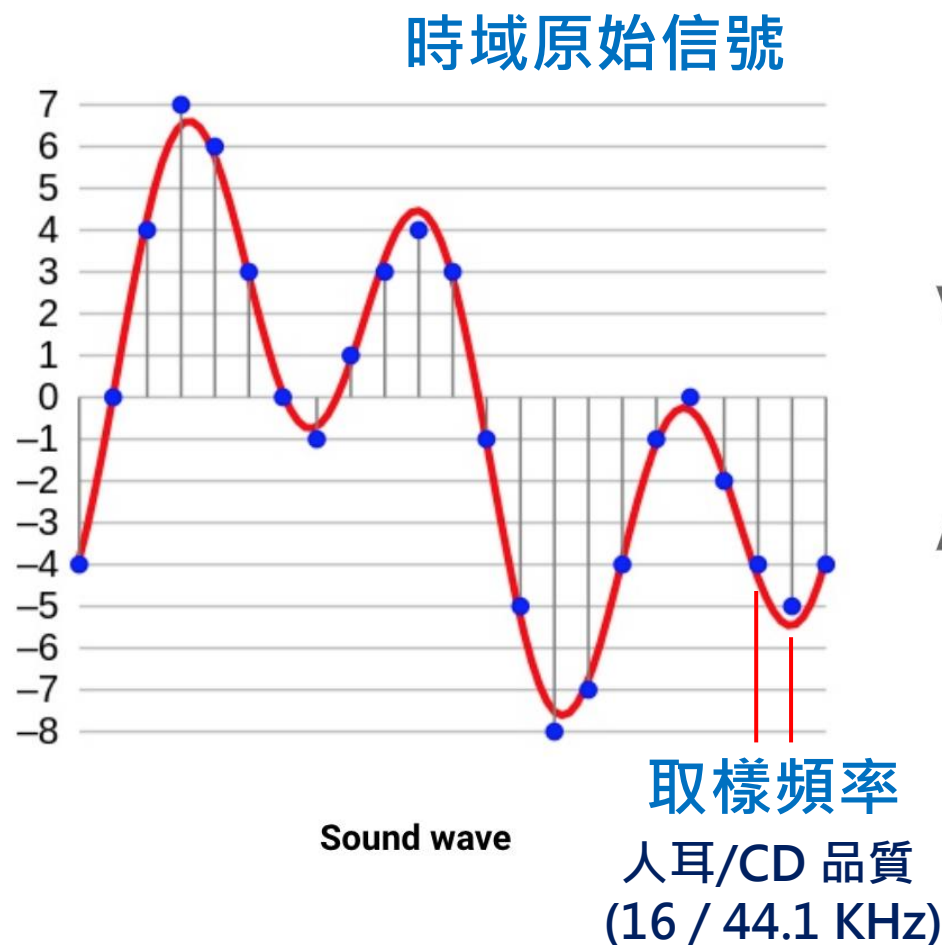


微機電式數位麥克風
(I2C, I2S 數位輸出)



手機、電腦麥克風
(檔案輸出)

聲音訊號—取樣、量化



量化數位資料
(量化位元數 8 / 12 / 16 bit)

[-4, 0, 4, 7, 6, 3, 0, -1, 1, 3, 4, 3,
-1, -5, -8, -7, -4, -1, 0, -2, -4, -5, -4]

資料大小 = 取樣頻率 * 量化位元數

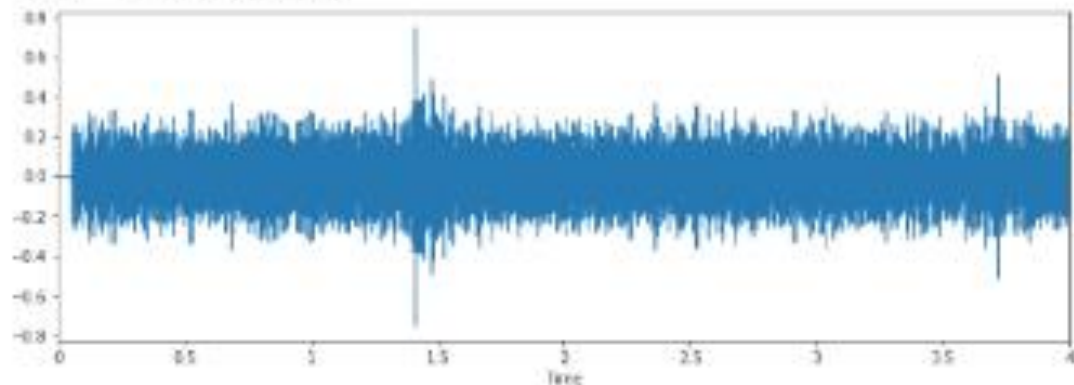
例：16KHz取樣率, 12bit, 取樣1秒
 $16 \text{ KHz} * 2 \text{ Byte} = 32 \text{ K Byte/Sec}$
 (9 ~16 bit $\rightarrow$ 2 Byte)

Array

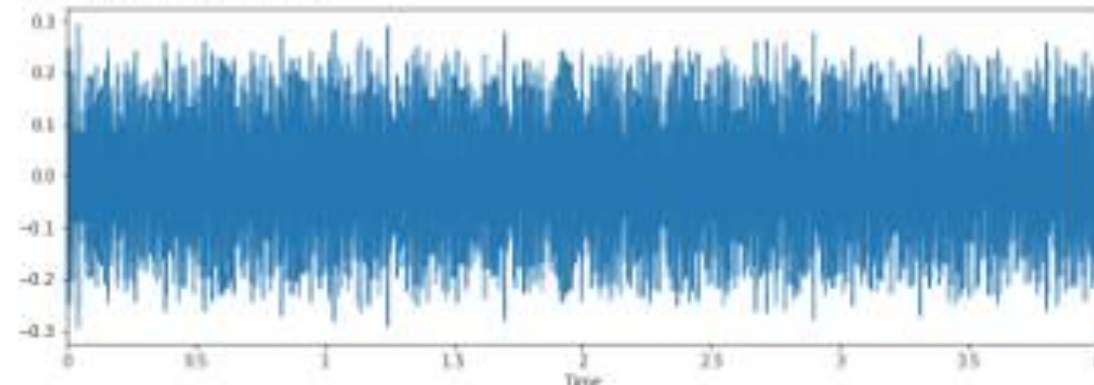
資料來源：<https://mikesmales.medium.com/sound-classification-using-deep-learning-8bc2aa1990b7>

常見聲音訊號波形（時間域）

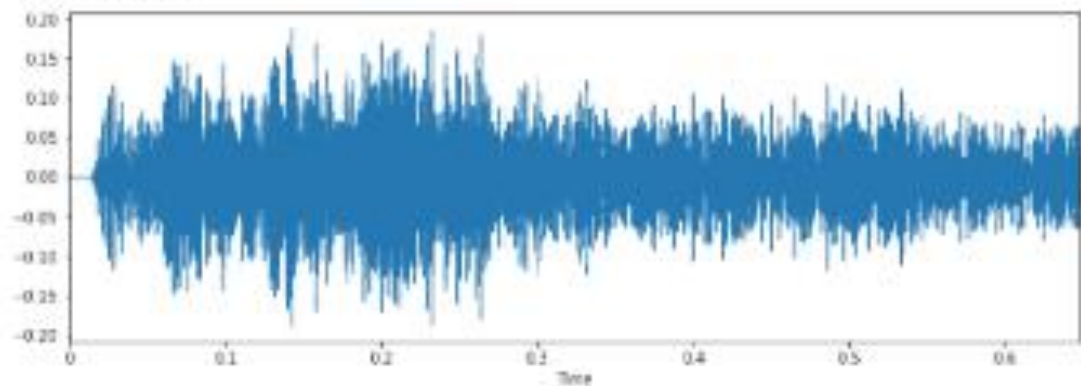
Air Conditioner



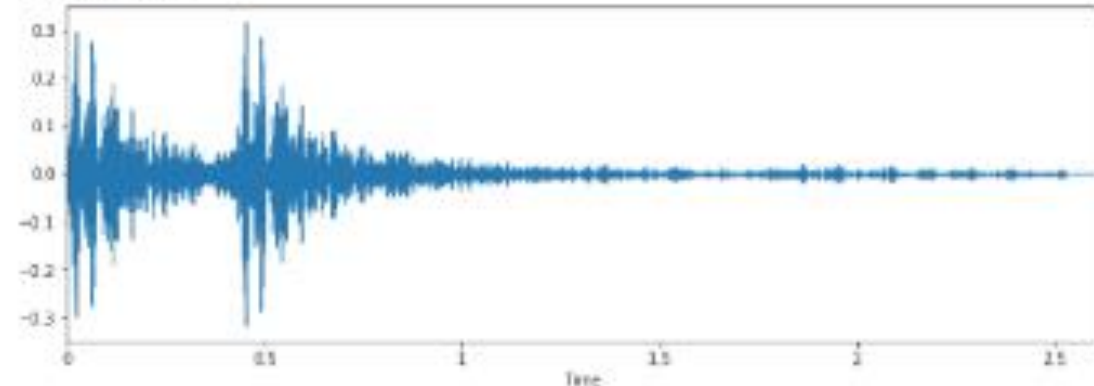
Engine Idling



Car horn



Gunshot

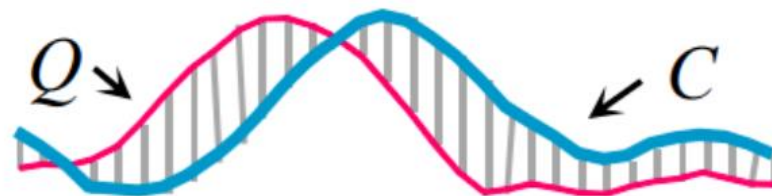


資料來源：<https://mikesmales.medium.com/sound-classification-using-deep-learning-8bc2aa1990b7>

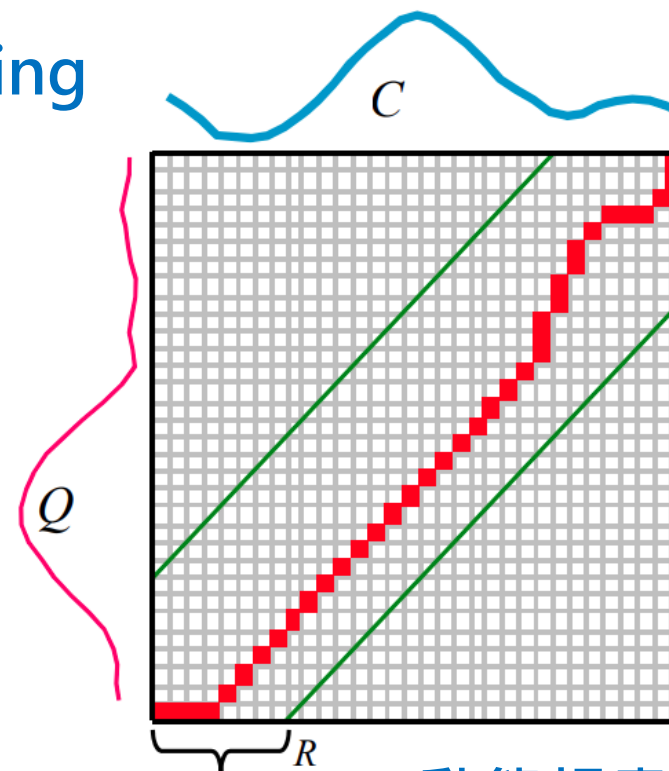
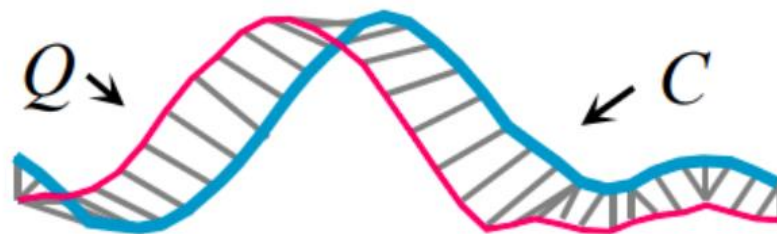
傳統時間序列相似度分析 — DTW

動態時間扭曲Dynamic time warping

時序
同步



時序
異步



動態規畫

歐氏
距離

$$d(x, y) = \sqrt{(x_1 - y_1)^2 + (x_2 - y_2)^2 + \cdots + (x_n - y_n)^2}$$

資料來源：https://en.wikipedia.org/wiki/Dynamic_time_warping

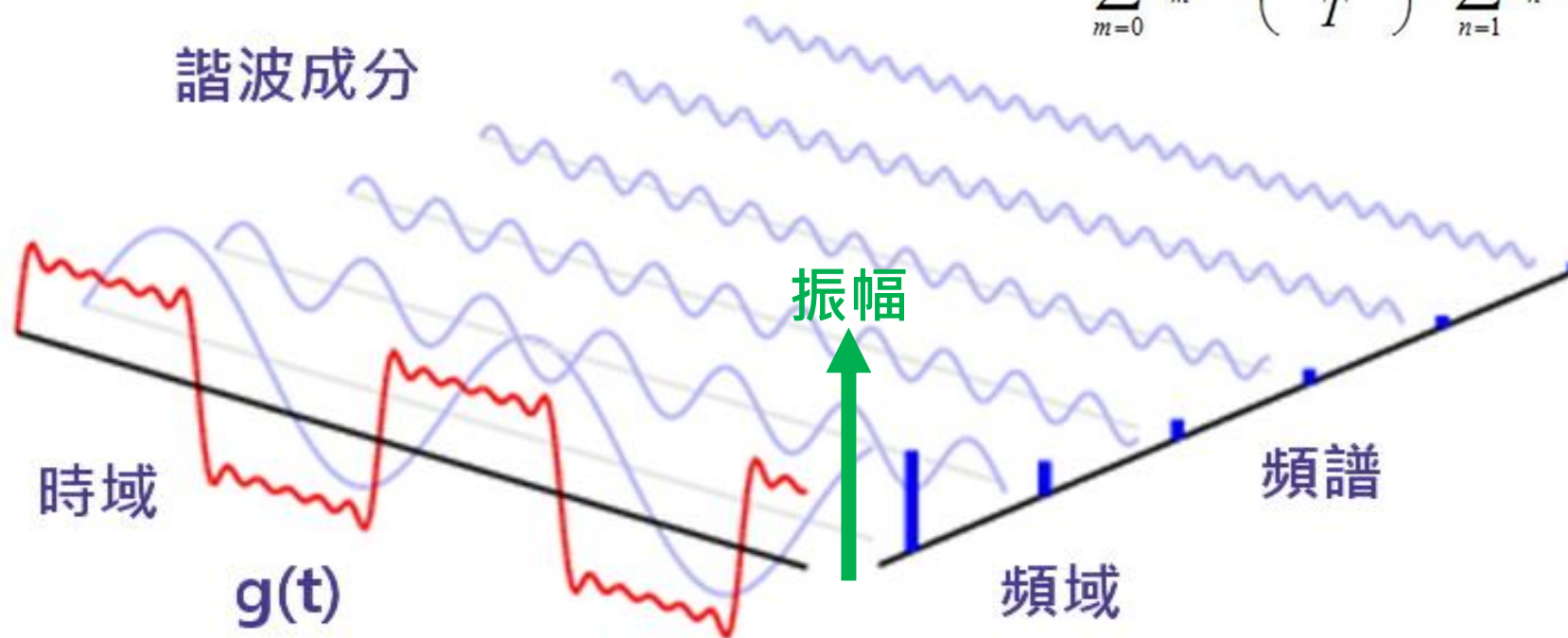
特徵提取 — 時間域、頻率域轉換

傅立葉級數（轉換）

所有信號都可拆解為不同頻率弦波之組合

$$g(t) = a_0 + \sum_{m=1}^{\infty} a_m \cos\left(\frac{2\pi mt}{T}\right) + \sum_{n=1}^{\infty} b_n \sin\left(\frac{2\pi nt}{T}\right)$$

$$= \sum_{m=0}^{\infty} a_m \cos\left(\frac{2\pi mt}{T}\right) + \sum_{n=1}^{\infty} b_n \sin\left(\frac{2\pi nt}{T}\right)$$

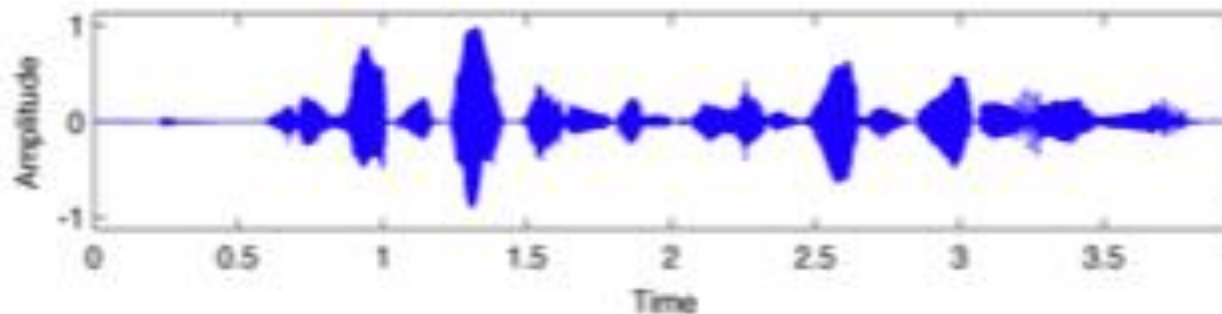


連續型？
間歇型？

資料來源：https://yhhuang1966.blogspot.com/2019/12/praat_5.html

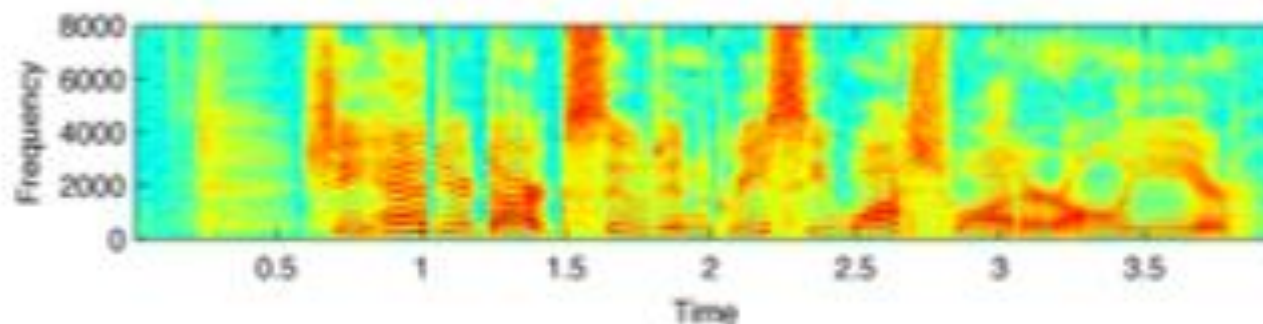
特徵提取 — 線性頻譜、梅爾倒頻譜表示法

**Time Domain
Waveform**



線性
頻譜

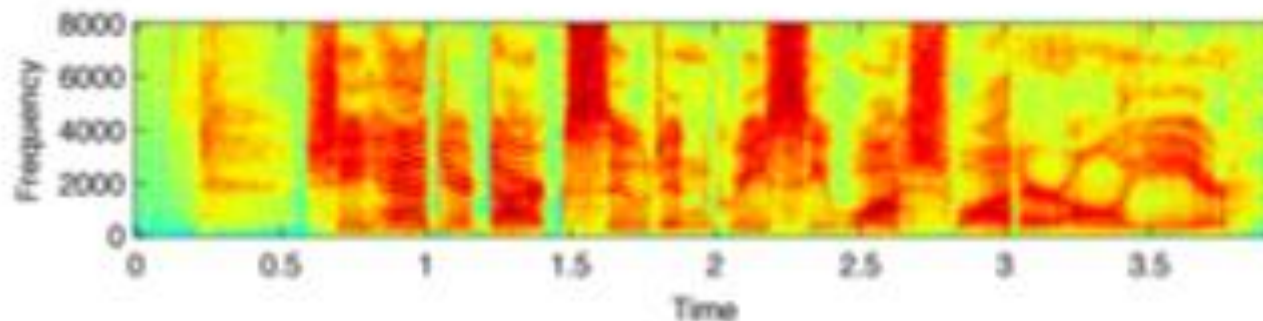
Spectrogram



對數
頻譜

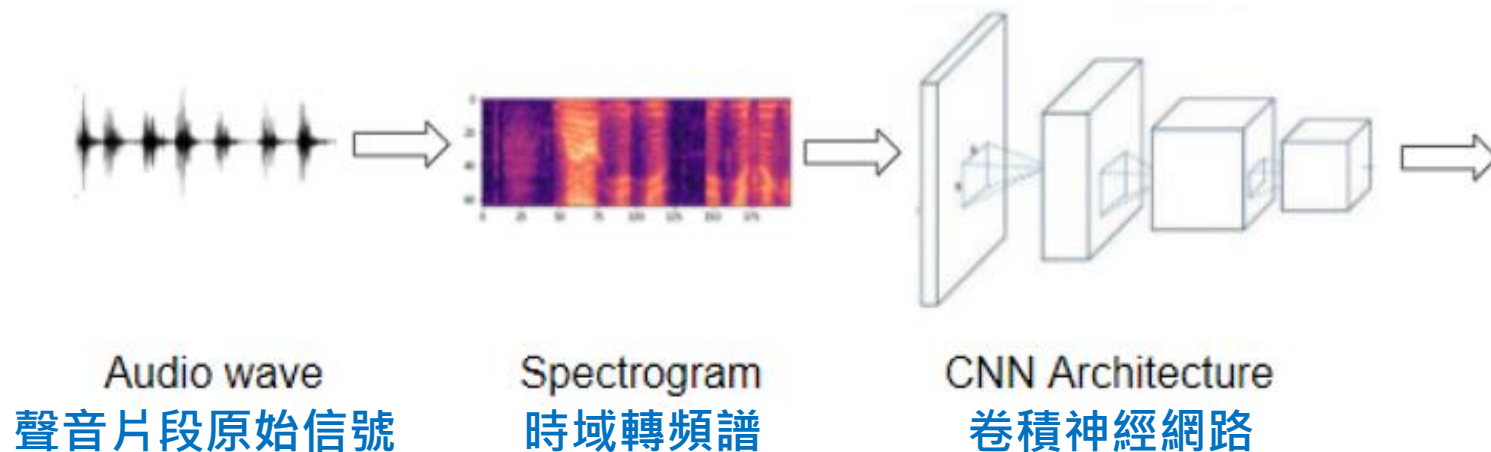
**MFCC
Spectrogram**

Mel-Frequency Cepstral
Coefficients (MFCC)

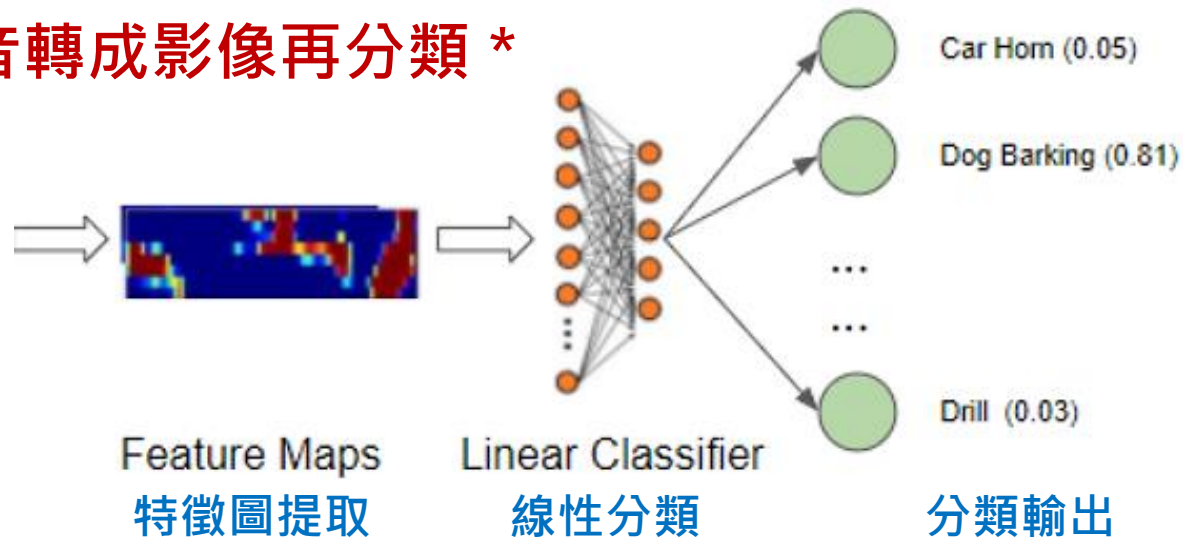


資料來源：<https://mikesmales.medium.com/sound-classification-using-deep-learning-8bc2aa1990b7>

深度學習模型聲音分類

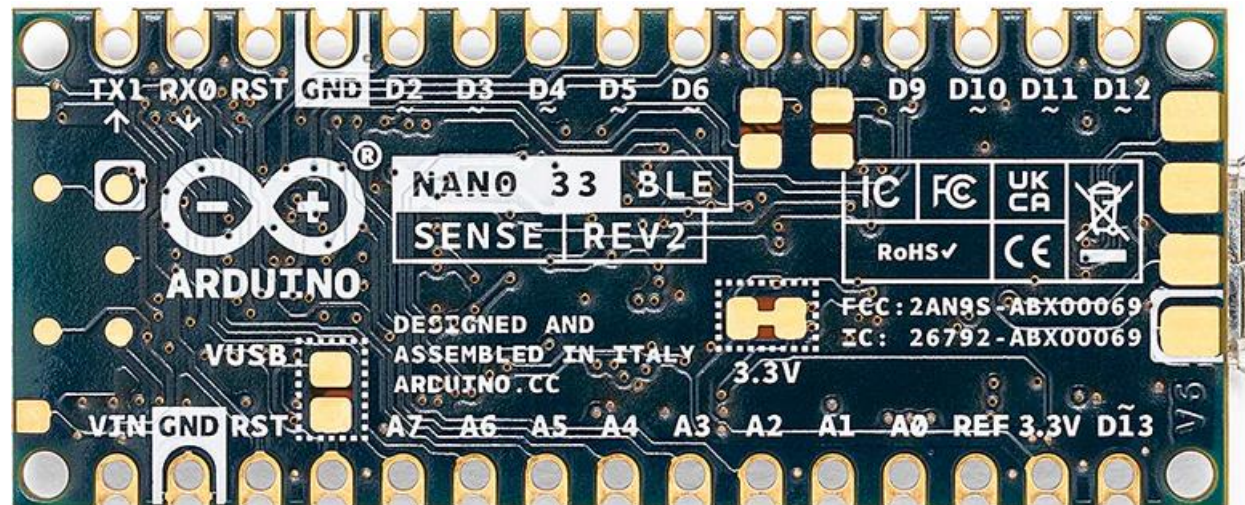
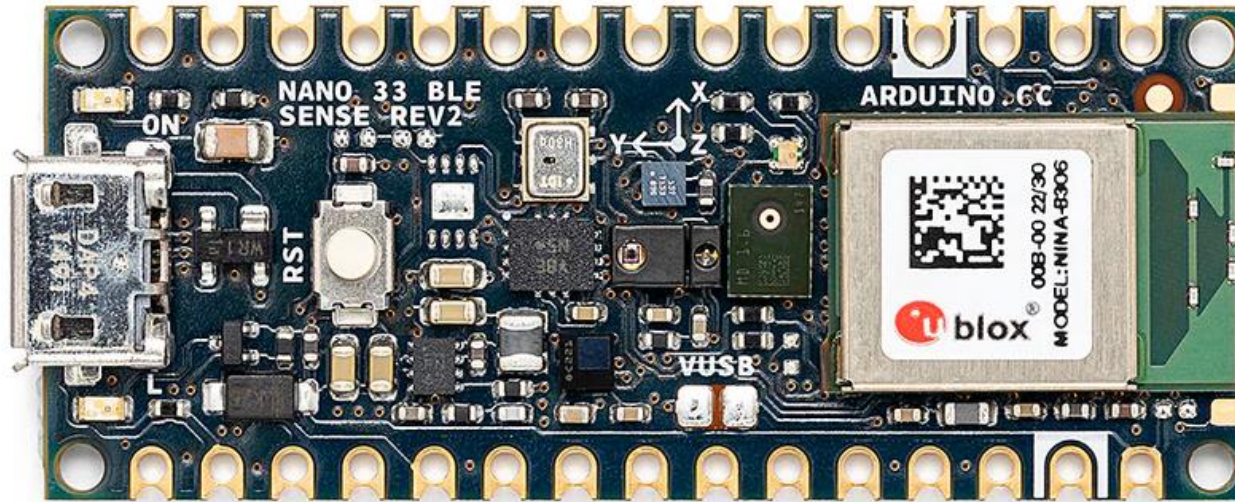


*** 把聲音轉成影像再分類 ***



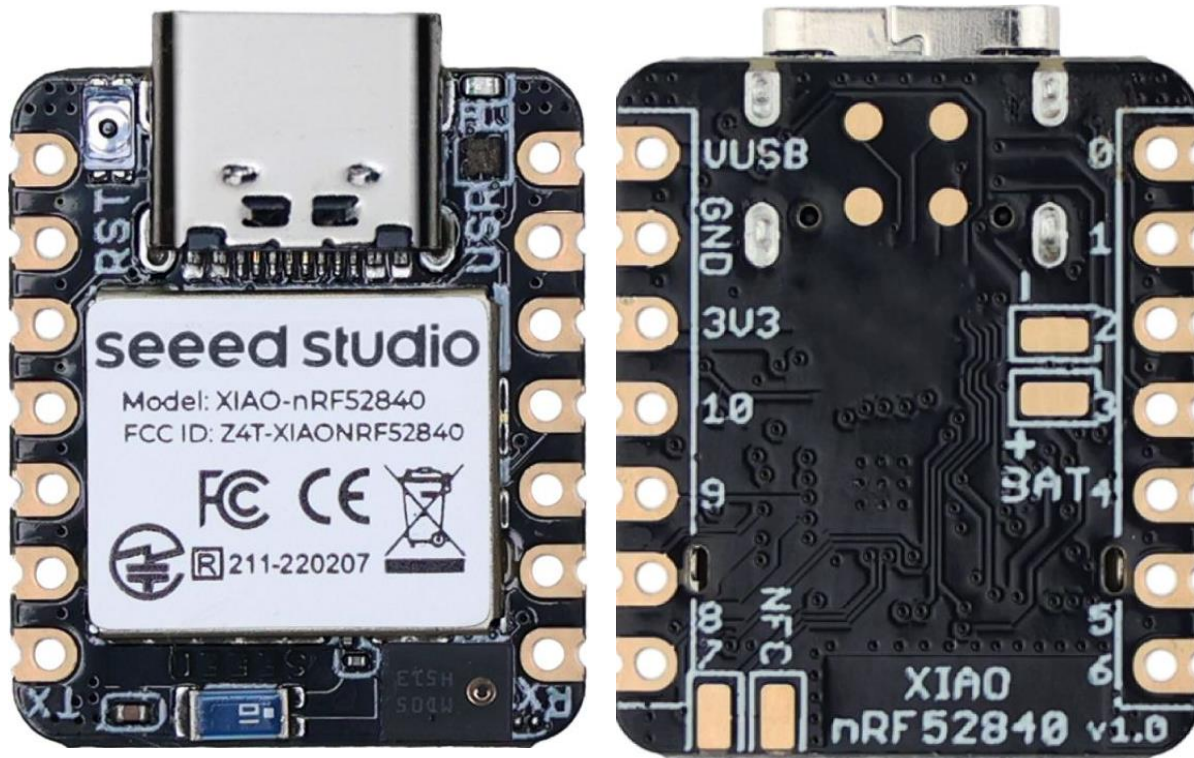
資料來源：<https://towardsdatascience.com/audio-deep-learning-made-simple-sound-classification-step-by-step-cebc936bbe5>

Arduino Nano 33 BLE Sense Rev.2 規格



- 微處理器：Nordic nRF52840 (arm Cortex-M4F)
- 工作電壓/頻率：3.3V / 64MHz
- **Flash / SRAM**：1MB / 256KB
- 外觀尺寸：45mm x 18mm
- 數位 IO：x14 (+板上LED)
- 類比輸入：12bit, 200KHz x8
- 類比輸出：PWM x14
- 感測器：
 - 運動感測器 (BMI270 + BMM150)
 - 數位麥克風(MP34DT06JTR)
 - 手勢/照度/近接/色彩(APDS9960)
 - 氣壓(LPS22HB)
 - 溫濕度(HS3003)

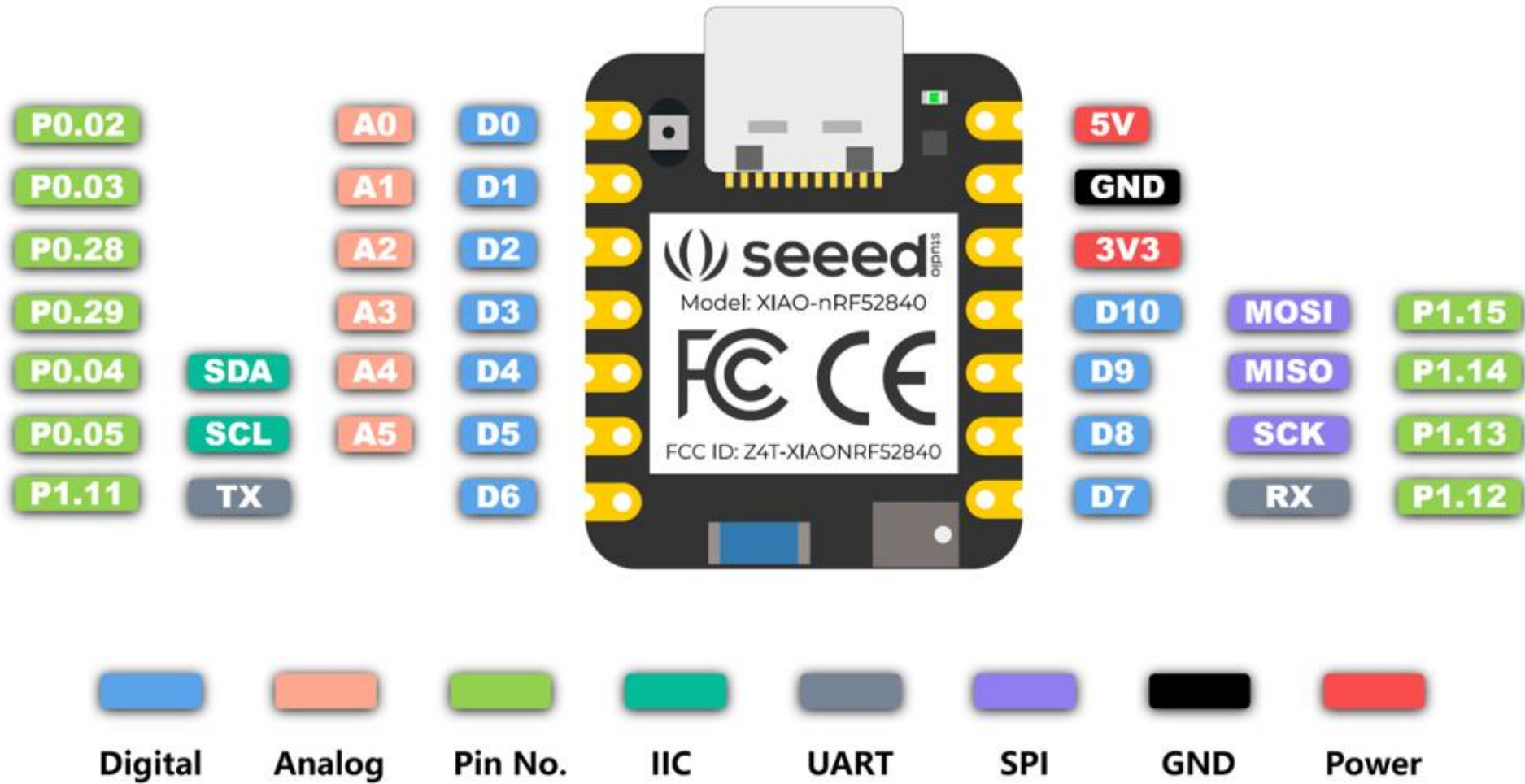
Seeed Xiao nRF52840 (Sense) 規格



資料來源：https://wiki.seeedstudio.com/XIAO_BLE/

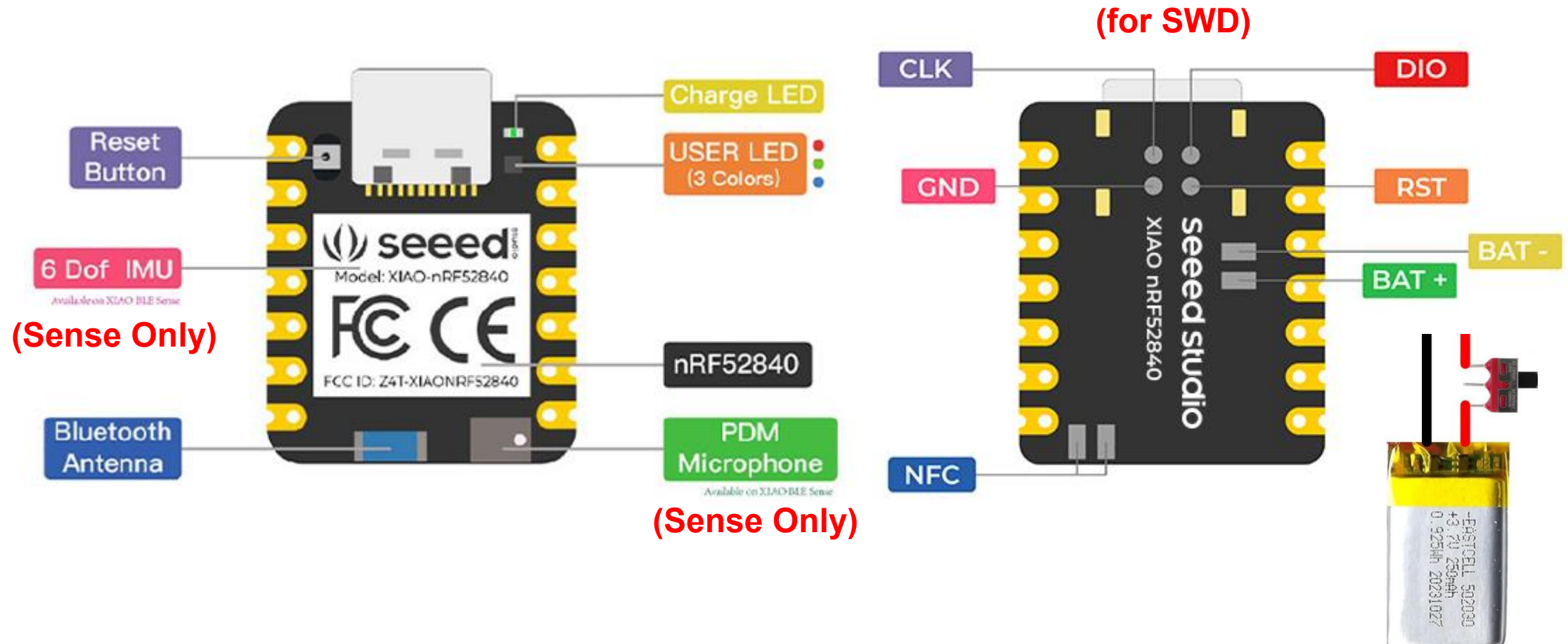
- 微處理器：
 - Nordic nRF52840 (arm Cortex-M4F)
- 工作電壓/頻率：3.3V / 64MHz
- **Flash / SRAM**：1MB / 256KB
- **外部Flash**：QSPI 2MB
- 外觀尺寸：21mm x 17.8mm
- 數位 IO：x11 (+板上RGBLED)
- 類比輸入 / 輸出：x6 / PWM x11
- 感測器：**(Sense Only)**
 - 運動感測器 (LSM6DS3TR-C)
 - 數位麥克風
- 無線：Bluetooth 5.0 / BLE / NFC
- 開發工具：
 - Arduino / MicroPython / CircuitPython

Seeed Xiao nRF52840 (Sense) 接腳圖



資料來源：https://wiki.seeedstudio.com/XIAO_BLE/

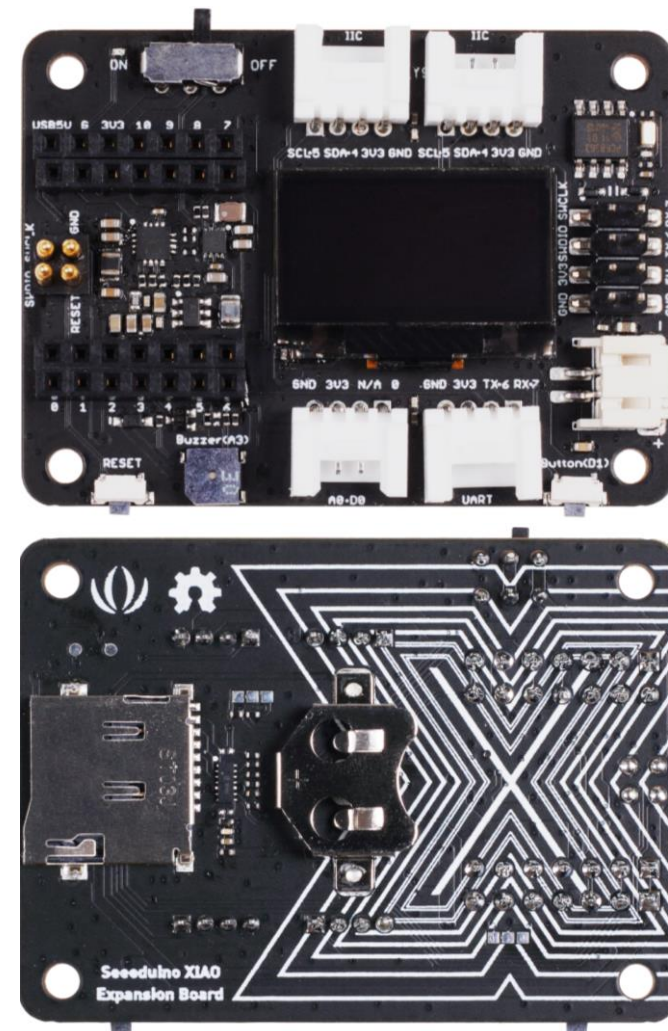
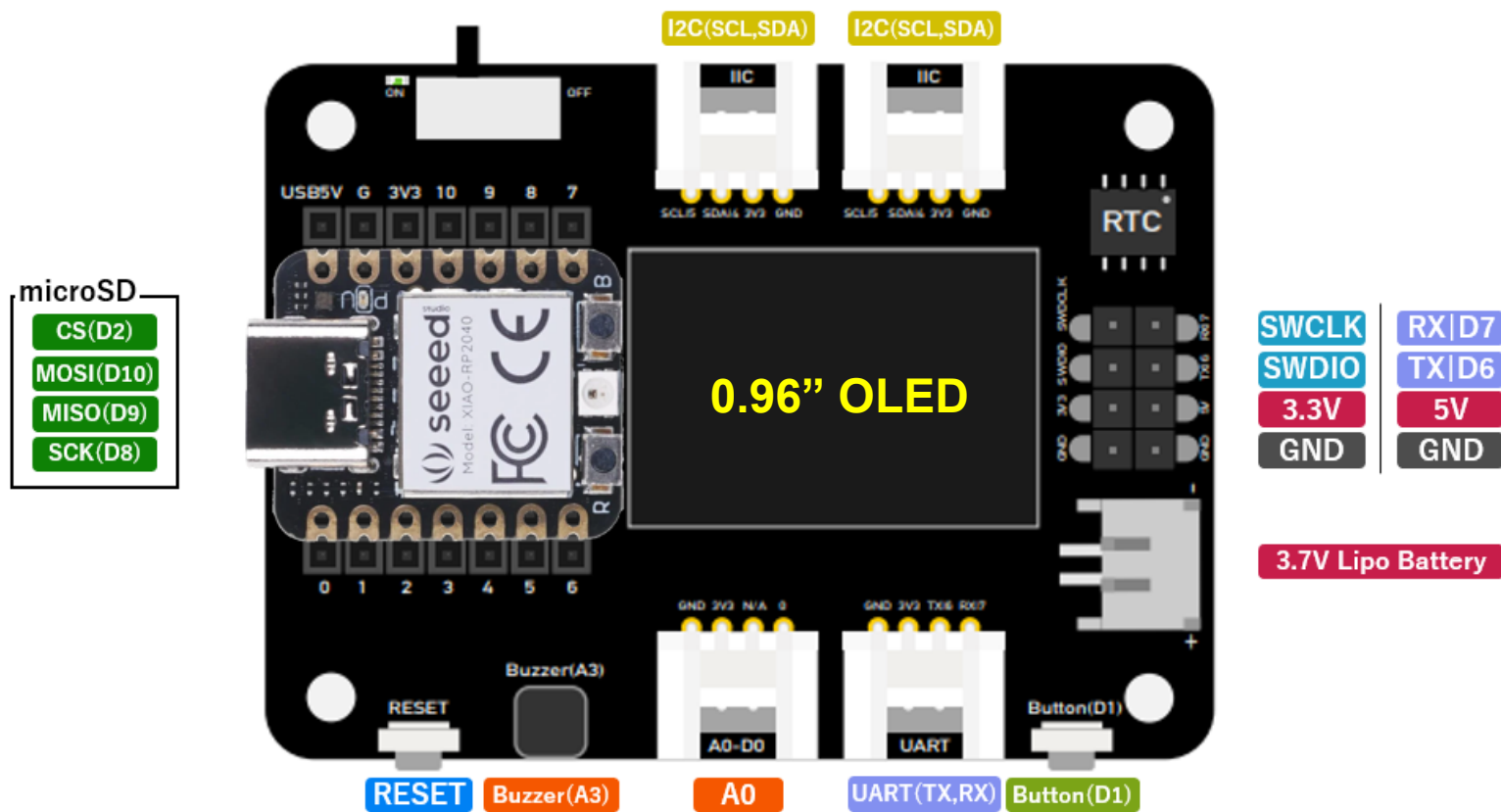
Seeed Xiao nRF52840 (Sense) 元件圖



資料來源：https://wiki.seeedstudio.com/XIAO_BLE/

Seeed Xiao 擴充底板

可支援 Xiao SAMD21, RP2040, nRF52840(Sense), ESP32C3, ESP32S3(Sense), ESP32C6, RP2350, RA4M1, MG24(Sense)



資料來源：<https://wiki.seeedstudio.com/Seeeduino-XIAO-Expansion-Board/>

Seeed Xiao nRF52840 (Sense) 技術文件

seeed studio Quick Links ▾ Explore with Topics ▾ FAQs ▾ Rangers ▾ Bazaar 🛒 AI Bot 🤖 SenseCraft AI 🧠 🔄 🔍 Search CTRL K

Exhibition for XIAO Series

- XIAO SAMD21 >
- XIAO RA4M1 >
- XIAO MG24 >
- XIAO RP2040 >
- XIAO RP2350 >
- XIAO nRF52840 Series ▾
- Getting Started with Seeed Studio XIAO nRF52840 Series
- RTOS >
- Programming Language >
- Platform >
- Hardware Usage ▾
- IMU Usage for XIAO nRF52840 Sense

Getting Started with Seeed Studio XIAO nRF52840 Series

XIAO nRF52840	XIAO nRF52840 Sense	XIAO nRF52840 Plus
		
Get One Now	Get One Now	Get One Now

Features

- Specifications comparison
- Hardware overview
- Two Arduino Libraries
- Getting started
 - Hardware setup
 - Software setup
- Playing with the built-in 3-in-one LED
- Power Consumption Verification
- Battery Charging current
- Access the SWD Pins for Debugging and Reflashing Bootloader
- FAQ
 - Q1: My Arduino IDE is stuck when uploading code to the board

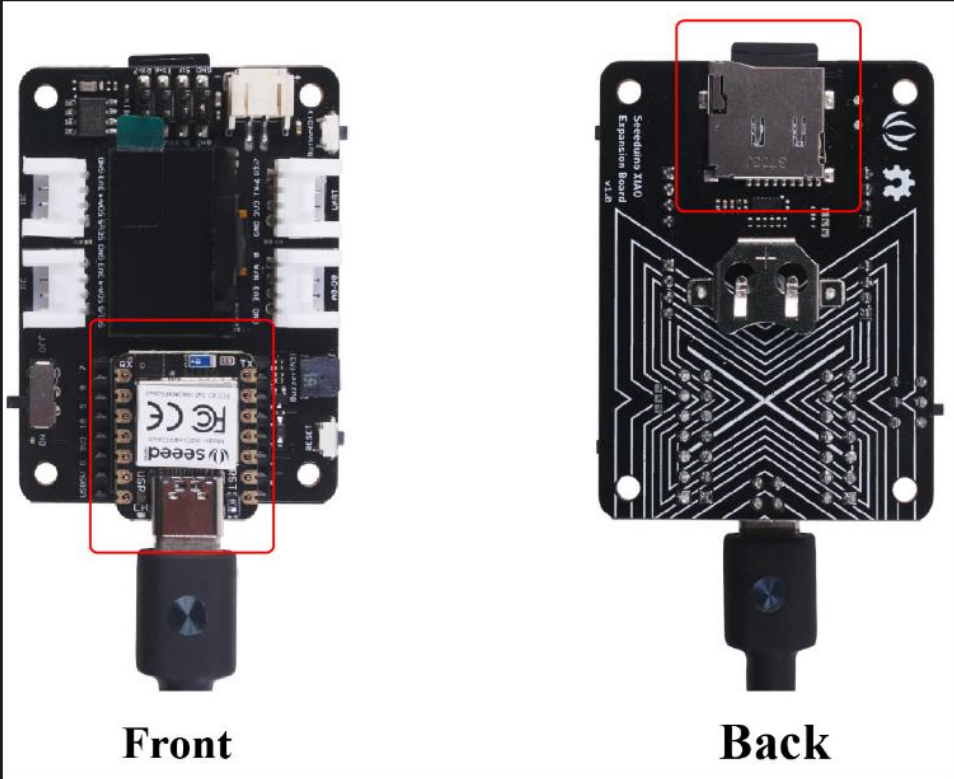
https://wiki.seeedstudio.com/XIAO_BLE/

Seeed Xiao nRF52840 Sense PDM

seeed studio Quick Links Explore with Topics FAQs Get Involved us English Bazaar SenseCraft AI

Programming Language
Platform
Hardware Usage
IMU Usage for XIAO nRF52840 Sense
PDM Usage for XIAO nRF52840 Sense
The QSPI Flash for XIAO nRF52840 Sense
Pin Multiplexing for both versions
NFC Usage for both versions
Bluetooth Libraries
Embedded ML
Application

Expansion Board and insert an SD card into the SD card slot of the expansion board.

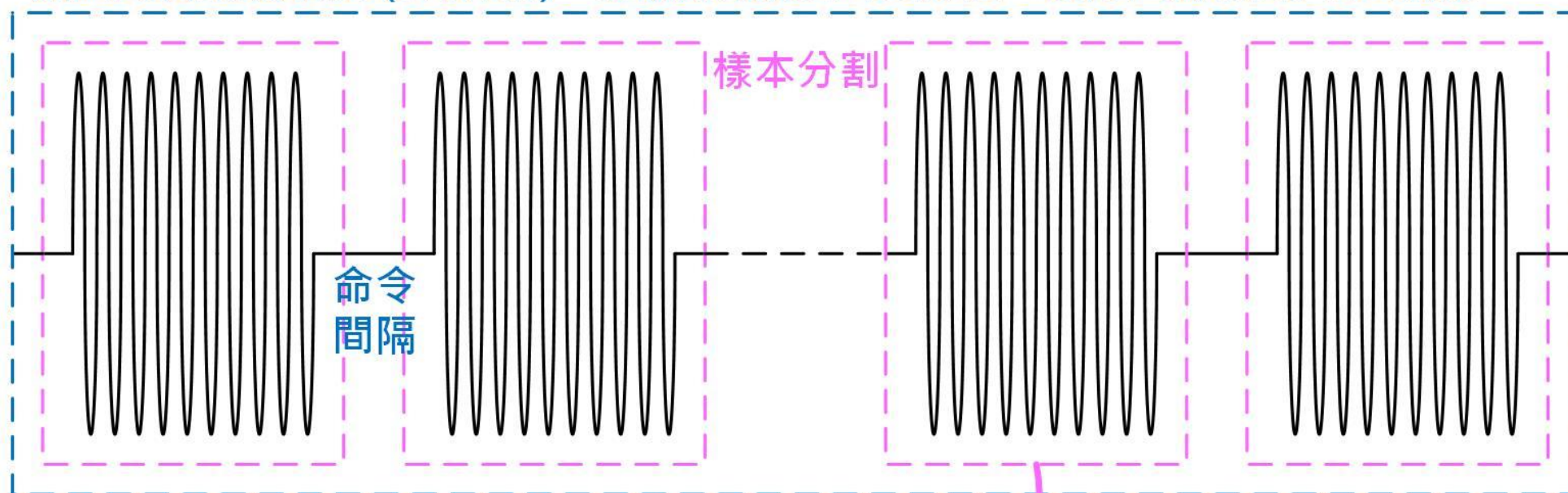


Front Back

<https://wiki.seeedstudio.com/XIAO-BLE-Sense-PDM-Usage/>

自定義語音命令錄音技巧

完整聲音錄製檔案 (on.wav) – 建議取樣頻率 16KHz，以非壓縮WAV格式儲存。



分割區間 1秒

完整聲音檔案分割
自建資料集方式

/on

on_001.wav

on_002.wav

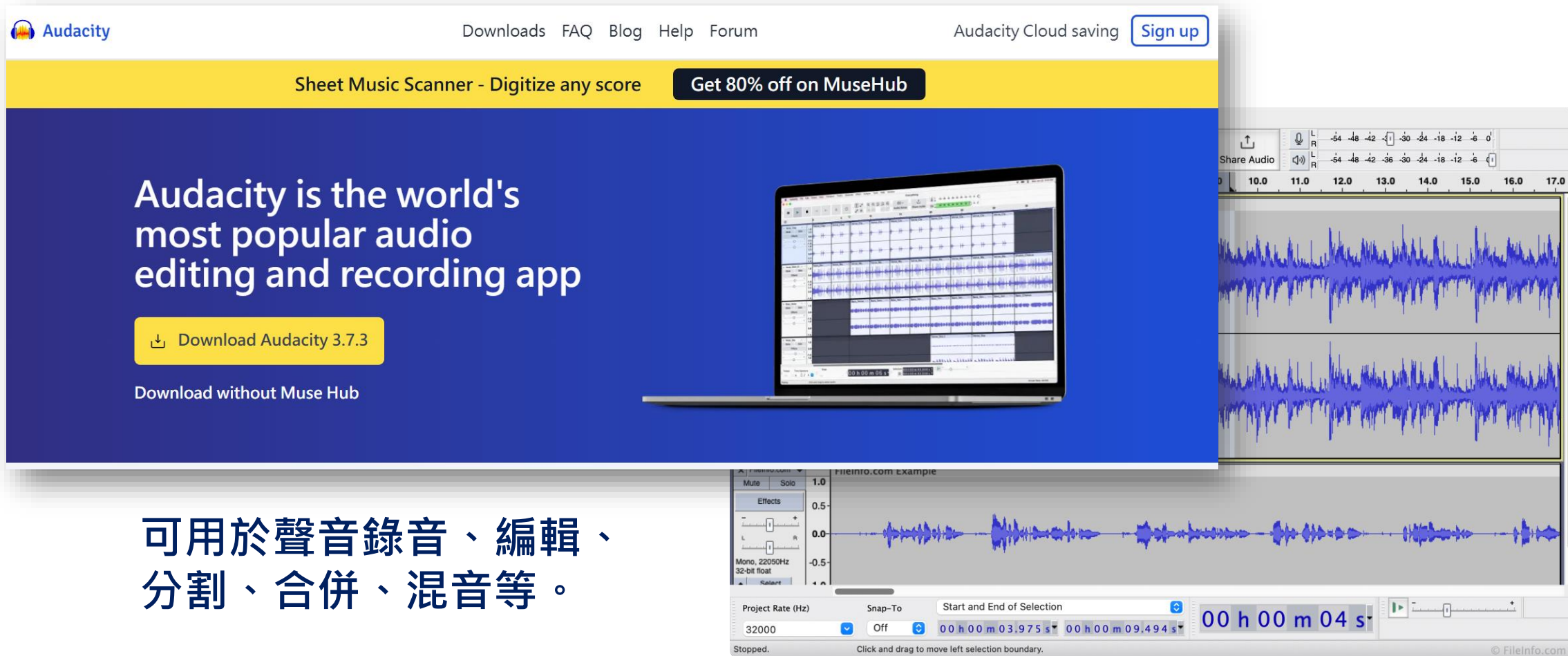
on_xxx.wav

另存新檔

OmniXRI整理繪製, 2021/9/30

資料來源：<https://github.com/OmniXRI/iThomeIronMan2021/blob/main/Day17.md>

聲音分割開源工具 Audacity



The image shows a composite of the Audacity website and the software interface. The website header includes the Audacity logo, navigation links (Downloads, FAQ, Blog, Help, Forum), and a 'Sign up' button for Audacity Cloud saving. A yellow banner promotes a 'Sheet Music Scanner' and a 'Get 80% off on MuseHub' offer. The main content area features the text 'Audacity is the world's most popular audio editing and recording app' and a 'Download Audacity 3.7.3' button. Below this, it says 'Download without Muse Hub'. To the right, a laptop displays the Audacity interface. The interface itself shows a multi-track audio project with waveforms. A detailed view of the waveform is shown on the right, with a time scale from 10.0 to 17.0 seconds. At the bottom, a zoomed-in view of the software's controls is shown, including the 'Effects' panel, 'Project Rate (Hz)' set to 32000, 'Snap-To' set to 'Off', and a time display showing '00 h 00 m 03.975 s' to '00 h 00 m 09.494 s'.

Audacity

Downloads FAQ Blog Help Forum

Audacity Cloud saving [Sign up](#)

Sheet Music Scanner - Digitize any score [Get 80% off on MuseHub](#)

Audacity is the world's most popular audio editing and recording app

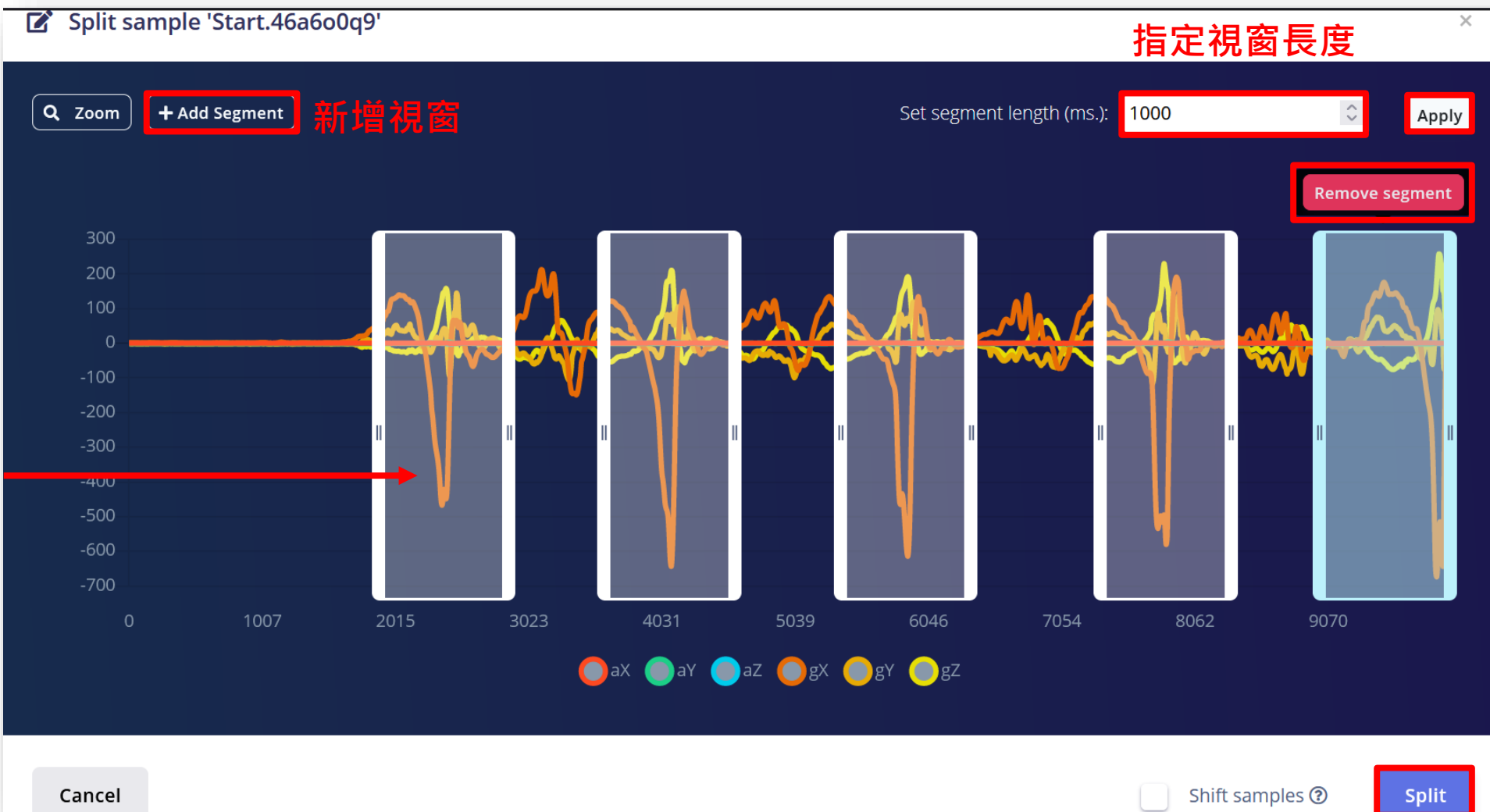
[Download Audacity 3.7.3](#)

Download without Muse Hub

可用於聲音錄音、編輯、分割、合併、混音等。

<https://www.audacityteam.org/>

Edge Impulse 連續信號自動分割工具

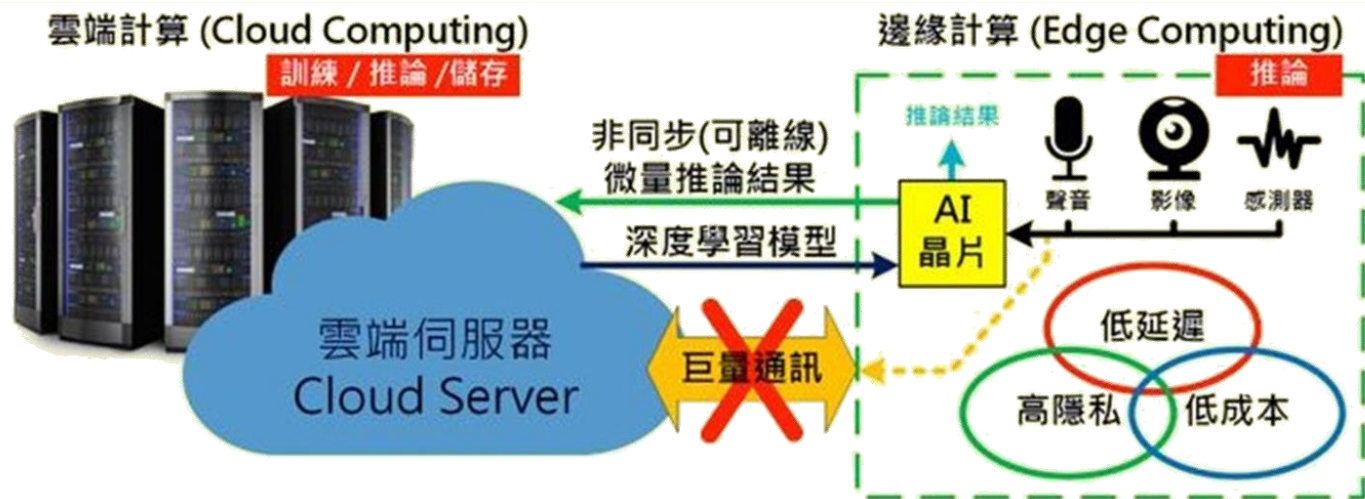


自動分割
刪除視窗

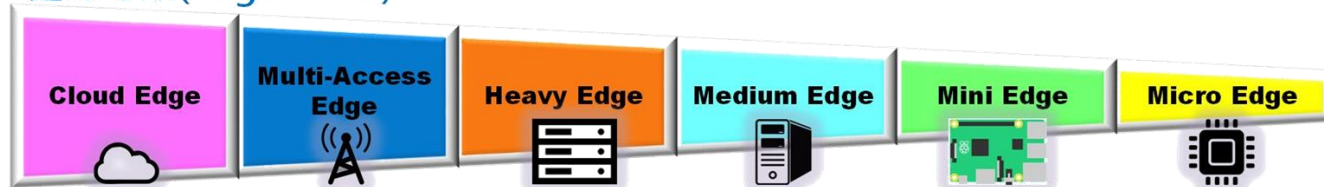
完成分割



Edge AI



邊緣等級(Edge Level)



8.2. 喚醒詞偵測

喚醒詞偵測

對一小段（數秒）聲音（連續信號）進行分類（辨識）如環境聲音（動物、機械、自然等）、生理訊號、機械振動、感測器變化等。

可以是非語言人聲或語音關鍵詞（非句子）。

常見同義詞

- **Keyword Spotting (KWS)**：語音關鍵詞定位
- **Keyword Detection**：語音關鍵詞偵測
- **Voice Command**：語音命令
- **Wake-Up-Word**：喚醒詞

常見應用

- 智能音箱、動作辨識、生理監看等



案例分享 — 聲控電風扇

正確示範（美女版）



<https://www.youtube.com/shorts/X8yJjRj7Uus>

錯誤示範（阿媽版）



<https://www.youtube.com/shorts/OdPn091H-0E>

正確示範（台灣國語）



<https://youtu.be/vpZgt0VQf8Y>

Google Speech Commands 公開資料集

共有常見語音命令35種，每筆資料1秒，包含數字、移動方向、常見動物及人名。曾多次釋出不同版本 v0.0.1, v0.0.2, v0.0.3，2 ~ 3GB。

資料集包含不同性別、各種口音及速度因此很適合用於一般智慧家庭應用。

實驗時為令訓練時間縮短可擷取較少數量進行訓練，待模型推論精度不足後再提升數量，以免浪費太多時間。

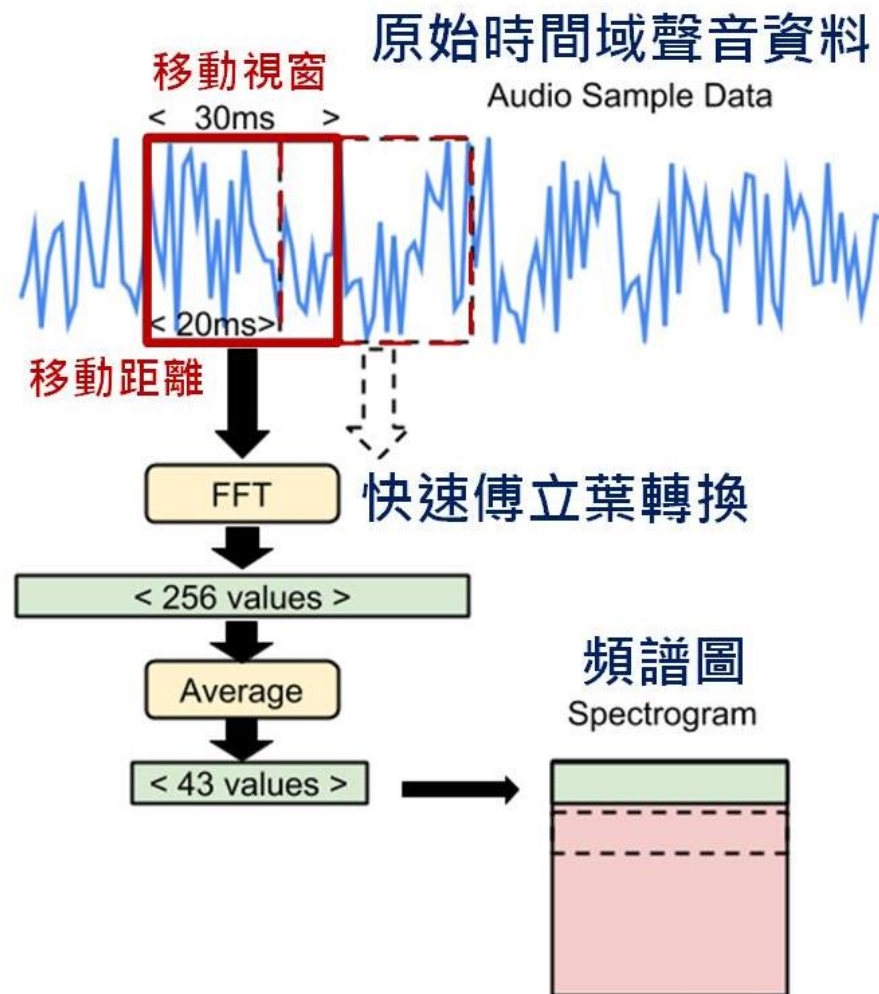
https://www.tensorflow.org/datasets/catalog/speech_commands

Speech Commands: A Dataset for Limited-Vocabulary Speech Recognition

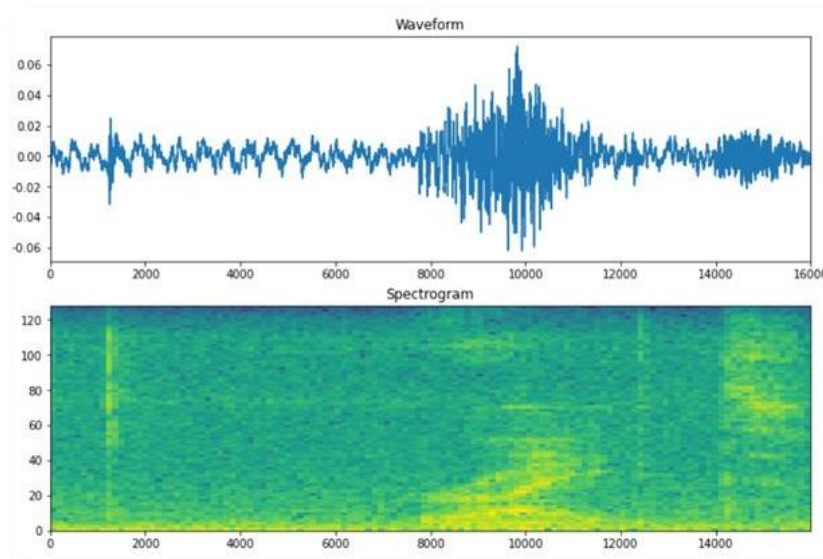
Word	Number of Utterances		
Backward	1,664	Nine	3,934
Bed	2,014	No	3,941
Bird	2,064	Off	3,745
Cat	2,031	On	3,845
Dog	2,128	One	3,890
Down	3,917	Right	3,778
Eight	3,787	Seven	3,998
Five	4,052	Sheila	2,022
Follow	1,579	Six	3,860
Forward	1,557	Stop	3,872
Four	3,728	Three	3,727
Go	3,880	Tree	1,759
Happy	2,054	Two	3,880
House	2,113	Up	3,723
Learn	1,575	Visual	1,592
Left	3,801	Wow	2,123
Marvin	2,100	Yes	4,044
		Zero	4,052

資料來源：<https://arxiv.org/abs/1804.03209>

關鍵詞偵測(KWS) 特徵提取



TensorFlow Lite Micro
Micro Speech
聲音頻譜圖轉換範例



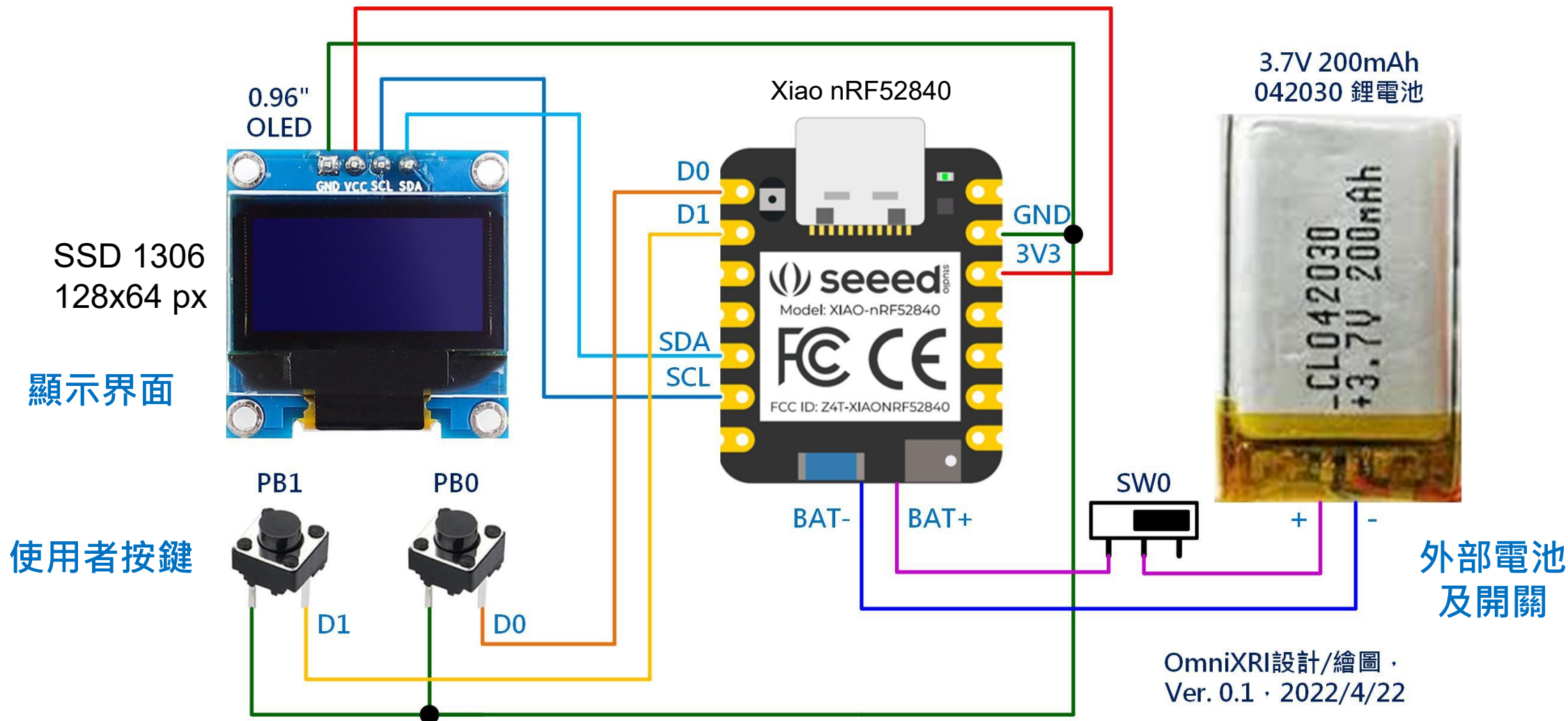
時間域格式

頻譜圖格式

資料來源：<https://github.com/OmniXRI/iThomeIronMan2021/blob/main/Day17.md>

OmniXRI整理繪製, 2021/9/30

Xiao nRF52840 Sense 模組 — 參考電路

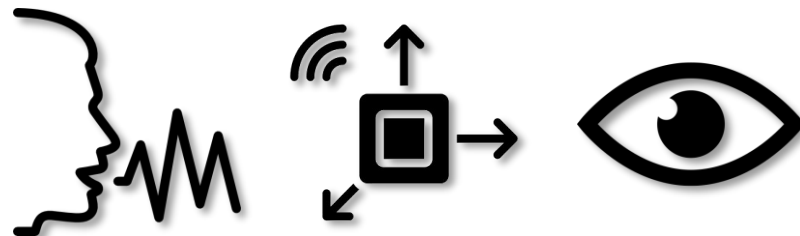


TinyML 開發流程選項

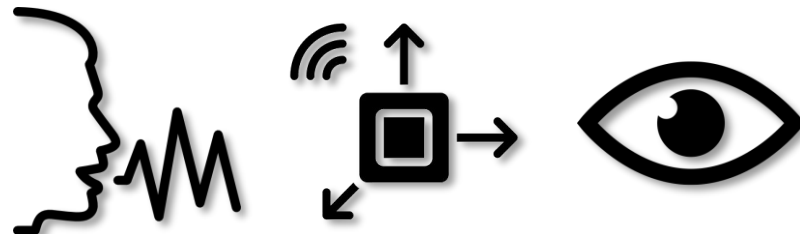
模型訓練工具名稱

適用感測器

主要特色



免費版：資料集需自行建置，僅提供模型訓練及轉成 TensorFlow Lite for Microcontroller 格式。可利用雲端 Google Colab 或本地端 Jupyter Notebooks 進行模型訓練及部署。



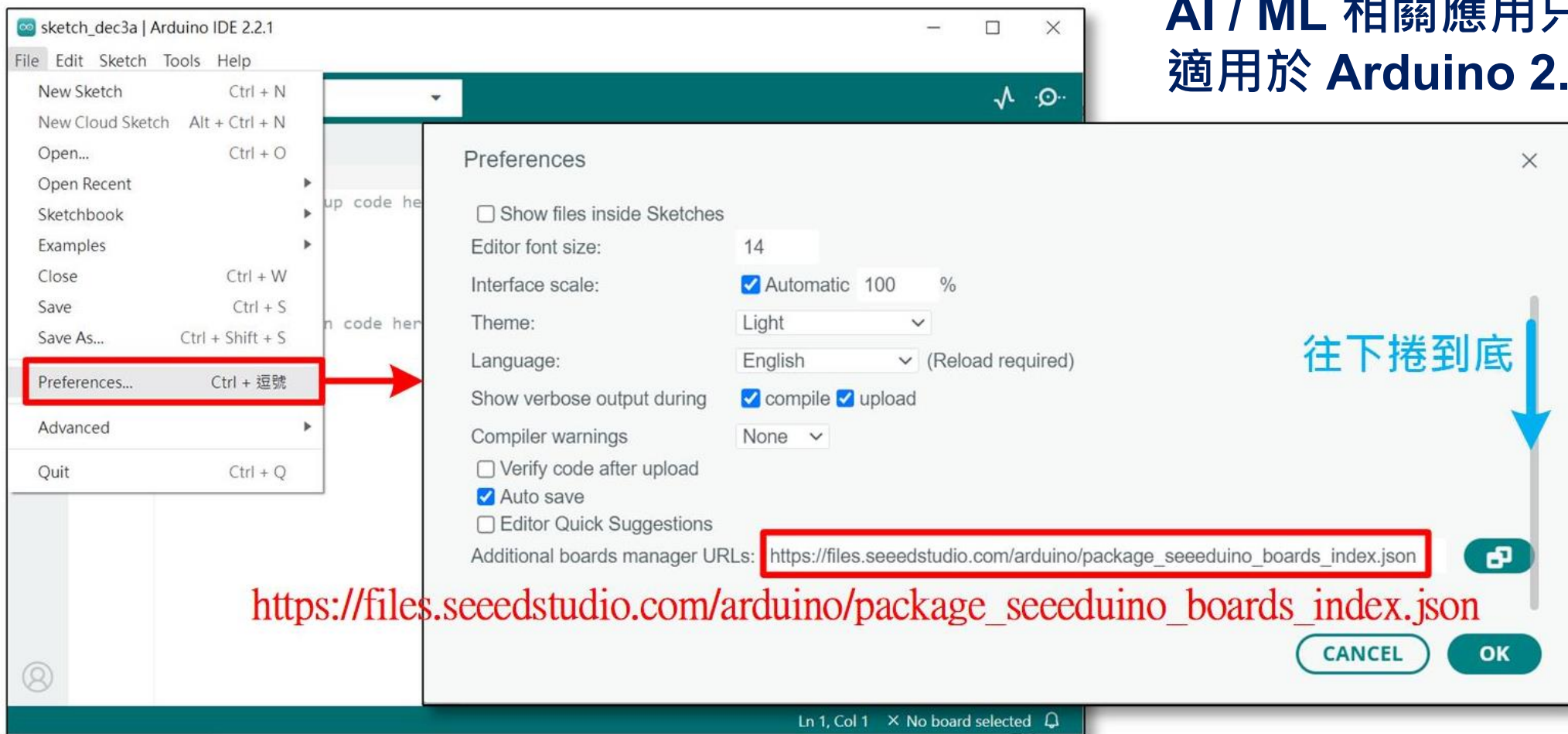
免費版：可支援各式感測器及開發板，提供一定額度資料集建置、資料前處理（特徵提取）模型訓練、模型優化及多種部署方式。亦可導入已訓練好的 TensorFlow Lite, ONNX 等模型。



免費版：影像分類可多分類、物件偵測同一模型僅能一類。事件觸發只能設一種條件。可支援雲端資料收集、模型訓練及部署，亦導入外部已訓練好之 TensorFlow Lite 模型。

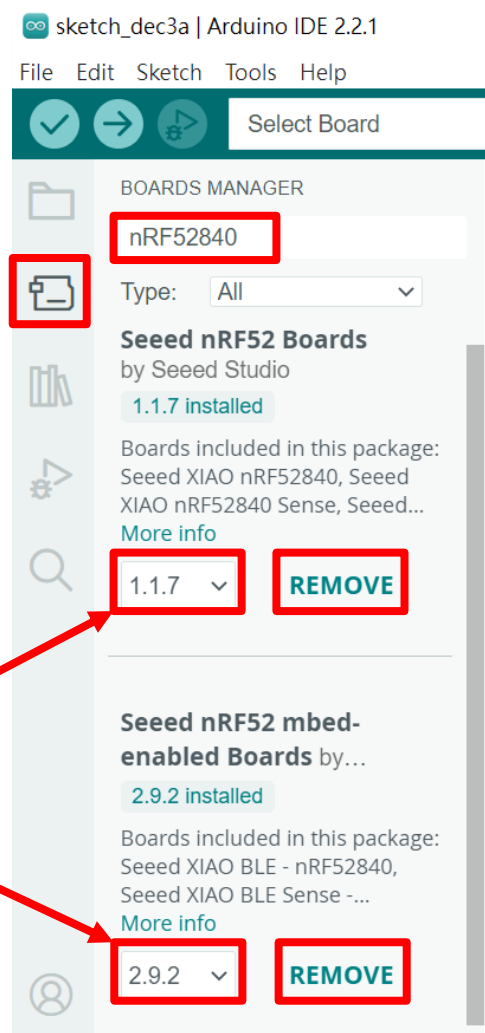
Arduino 新增開發板設定

AI / ML 相關應用只
適用於 Arduino 2.x 。



OmniXRI整理製作, 2023/12/08

安裝Seeed nRF52840函式庫



- 點選選單 Tools > Board > Boards Manager...，或直接點選左側第二個「開發板」圖示。
- 輸入 **nRF52840** 搜尋 Seeed nRF52840 開發板相關函式庫。
- 點選「**INSTALL**」安裝下列二個函式庫。（版本可取最新的）
 - **Seeed nRF52 Boards**（BLE, 低功耗功能）
 - **Seeed nRF52 mbed-enabled Boards**（PDM, IMU, ML）
- 安裝後若不需要時，可點選「**REMOVE**」解除安裝。

指定工作開發板及埠號

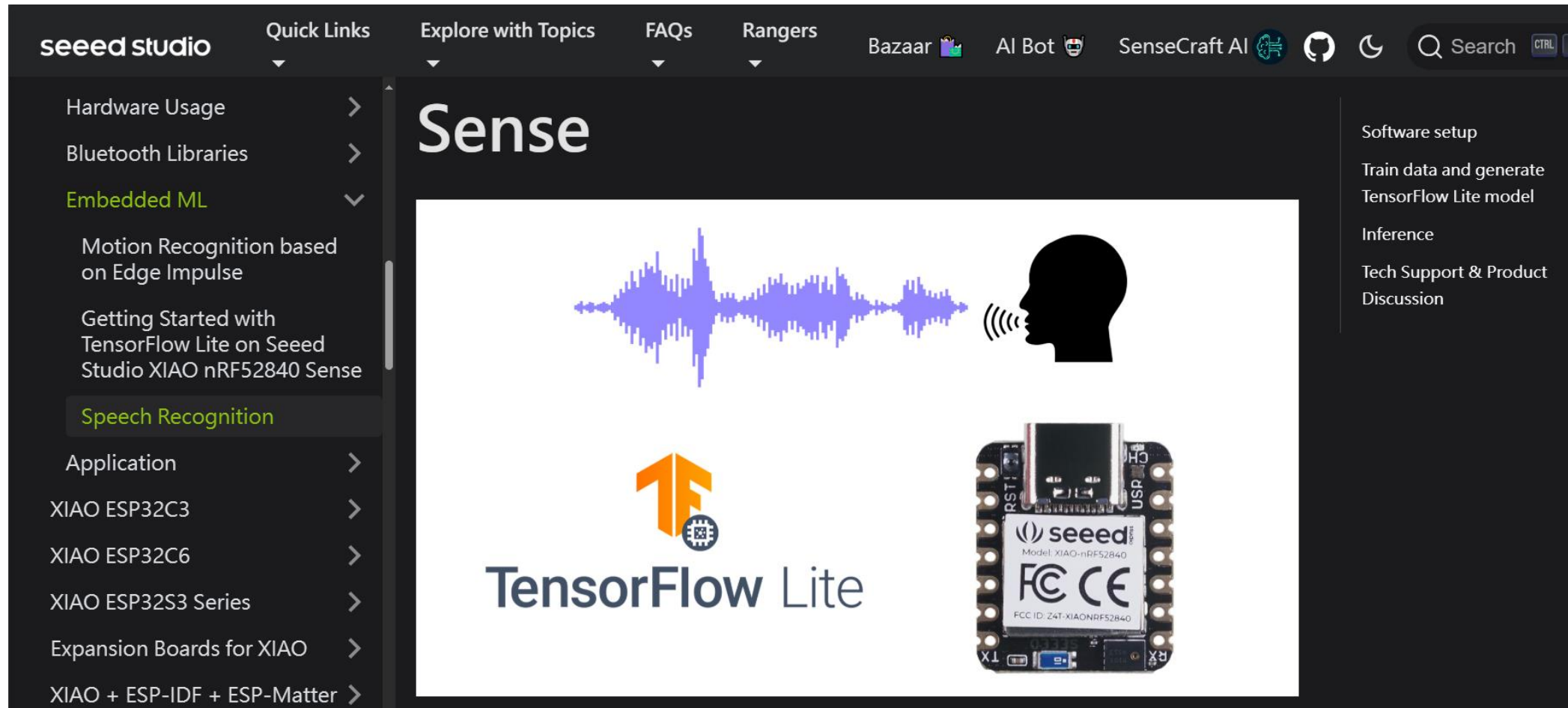
註：COM埠號每次插拔可能會不同

Ln 10, Col 1 Seeed XIAO nRF52840 on COM5

OmniXRI整理製作, 2023/12/08

- 選擇開發板
Seeed XIAO
nRF52840
Sense
- 選擇對應埠號
- 檢查是否連線

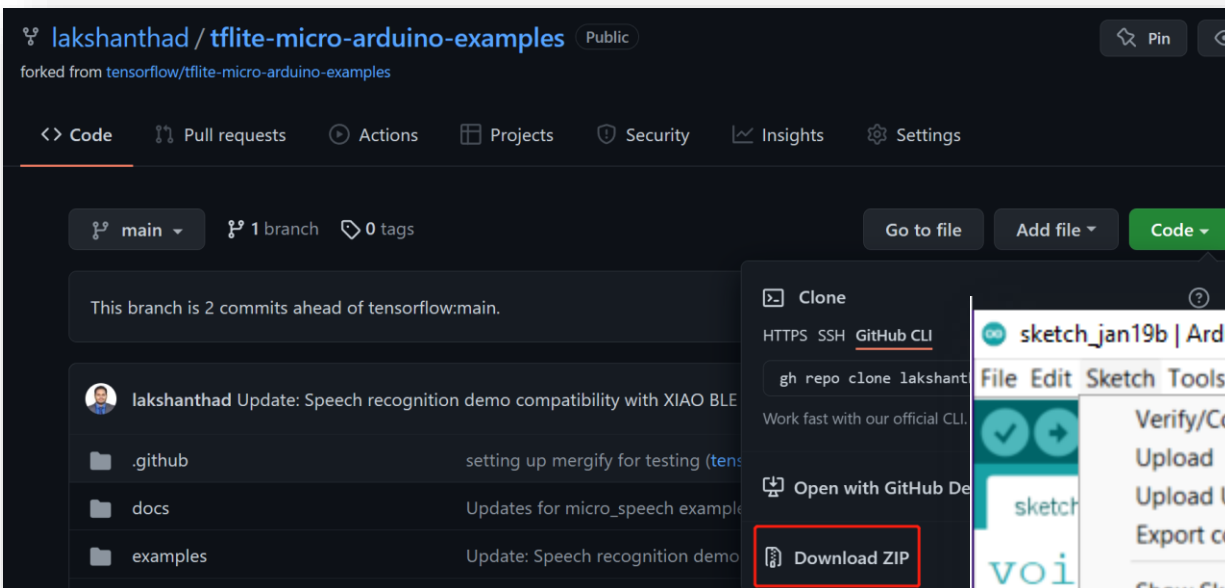
Xiao nRF52840 Sense 語音辨識



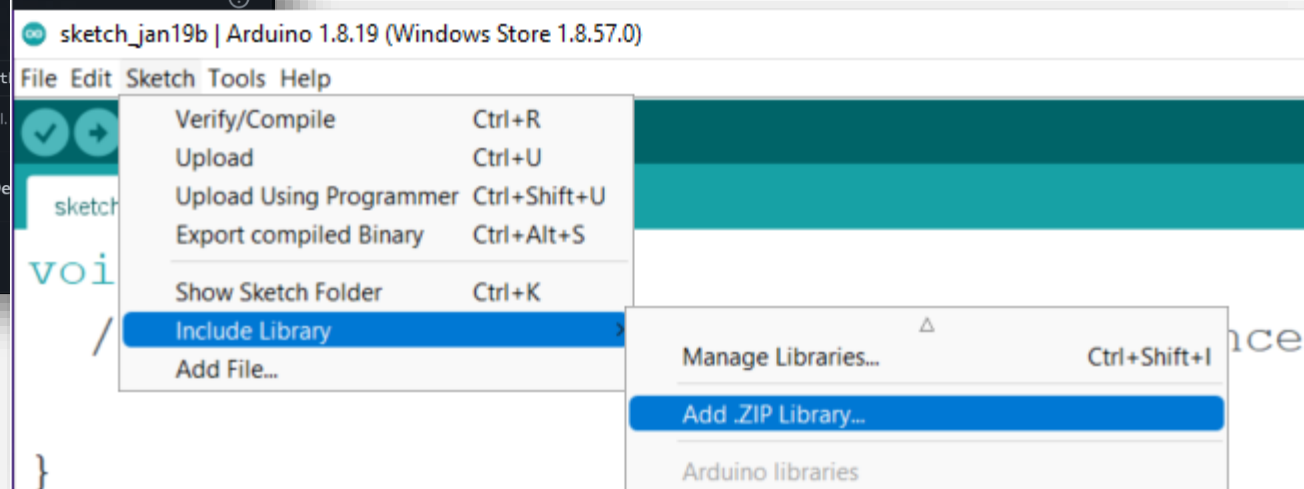
<https://wiki.seeedstudio.com/XIAO-BLE-Sense-TFLite-Mic/>

Aruino 預安裝模組

下載 **tflite-micro-arduino-examples** library (*.zip) 後，再安裝到 Arduino 函式庫中。(不用解壓縮)



<https://github.com/lakshanthad/tflite-micro-arduino-examples>



資料來源：<https://wiki.seeedstudio.com/XIAO-BLE-Sense-TFLite-Mic/>

啟動 Google Colab 模型訓練範例

The screenshot shows a Google Colab notebook interface. On the left, the file explorer displays a folder named 'sample_data' containing several files: 'README.md', 'anscombe.json', 'california_housing_test.csv', 'california_housing_train.csv', 'mnist_test.csv', and 'mnist_train_small.csv'. The main area of the notebook shows the title 'Train a Simple Audio Recognition Model' and a description: 'This notebook demonstrates how to train a 20 kB Simple Audio Recognition model to recognize keywords in speech. The model created in this notebook is used in the micro_speech example for TensorFlow Lite for MicroControllers.' Below this, there are links to 'Run in Google Colab' and 'View source on GitHub'. A section titled 'Configure Defaults' instructs the user to 'MODIFY the following constants for your specific use case.' The bottom status bar indicates '已連線至「Python 3 Google Compute Engine 後端 (GPU)」'.

https://colab.research.google.com/github/tensorflow/tflite-micro/blob/main/tensorflow/lite/micro/examples/micro_speech/train/train_micro_speech_model.ipynb

修正 Colab 範例訓練參數

✓ Configure Defaults

MODIFY the following constants for your specific use case.

挑選所需語音命令，一開始
可只選擇二種加速模型訓練

```
[ ] # A comma-delimited list of the words you want to train for
# The options are: yes,no,up,down,left,right,on,off,stop,go,
# All the other words will be used to train an "unknown" label and silent
# audio data with no spoken words will be used to train a "silence" label
WANTED_WORDS = "yes,no"

# The number of steps and learning rates can be specified as comma-separated
# lists to define the rate at each stage. For example,
# TRAINING_STEPS=12000,3000 and LEARNING_RATE=0.001,0.0001
# will run 12,000 training loops in total, with a rate of 0.001 for the first
# 8,000, and 0.0001 for the final 3,000.
TRAINING_STEPS = "12000,3000"
LEARNING_RATE = "0.001,0.0001"
```

"yes", "no", "up", "down", "left",
"right", "on", "off", "stop", "go",
"zero", "one", "two", "three", "four",
"five", "six", "seven", "eight", "nine",
"bed", "bird", "cat", "dog", "happy",
"house", "marvin", "sheila", "tree",
"wow"

訓練次數可降至1200, 300 加速模
型訓練，若推論效果不佳再往上加

Colab 開始執行模型訓練

The screenshot shows the Google Colab interface for a notebook titled 「train_micro_speech_model.ipynb」. The 'Run' menu is open, highlighting the '全部執行' (Run All) option with the keyboard shortcut Ctrl+F9. The menu also lists other options like '執行上方的儲存格' (Run cells above), '執行聚焦的儲存格' (Run focused cell), '執行選取範圍' (Run selection), '執行儲存格和下方所有儲存格' (Run cell and all cells below), '中斷執行' (Interrupt), '重新啟動工作階段' (Restart), '重新啟動工作階段並執行所有儲存格' (Restart and run all cells), '中斷連線並刪除執行階段' (Disconnect and delete), '變更執行階段類型' (Change execution type), '管理工作階段' (Manage execution), '查看資源' (View resources), and '查看執行階段記錄' (View execution history).

On the right side of the notebook, there is a text box with the following text:

訓練時長依設定參數可能一到數個小時
加速方式：

- 減少種類
- 減少每個種類樣本數
- 減少訓練次數

The background code in the notebook includes instructions for training a model, such as specifying words to train for, learning rates, and the number of loops.

Colab 完成模型訓練

▼ Deploy to a Microcontroller

Follow the instructions in the [micro_speech](#) README.md for [TensorFlow Lite for MicroControllers](#) to deploy this model on a specific microcontroller.

Reference Model: If you have not modified this notebook, you can follow the instructions as is, to deploy the model. Refer to the [micro_speech/train/models](#) directory to access the models generated in this notebook.

New Model: If you have generated a new model to identify different words: (i) Update `kCategoryCount` and `kCategoryLabels` in [micro_speech/micro_features/micro_model_settings.h](#) and (ii) Update the values assigned to the variables defined in [micro_speech/micro_features/model.cc](#) with values displayed after running the following cell.

```
# Print the C source file
!cat {MODEL_TFLITE_MICRO}

unsigned char g_model[] = {
    0x20, 0x00, 0x00, 0x00, 0x54, 0x46, 0x4c, 0x33, 0x00, 0x00, 0x00, 0x00,
    0x14, 0x00, 0x20, 0x00, 0x1c, 0x00, 0x18, 0x00, 0x14, 0x00, 0x10, 0x00,
    0x0c, 0x00, 0x00, 0x00, 0x08, 0x00, 0x04, 0x00, 0x14, 0x00, 0x00, 0x00,
    0x1c, 0x00, 0x00, 0x00, 0x1c, 0x00, 0x00, 0x00, 0x9c, 0x00, 0x00, 0x00,
    0x03, 0x00, 0x00, 0x00, 0x00, 0x00, 0x00, 0x03, 0x0c, 0x00, 0x0c, 0x00,
    0x0b, 0x00, 0x00, 0x00, 0x00, 0x00, 0x04, 0x00, 0x0c, 0x00, 0x00, 0x00,
    0x16, 0x00, 0x00, 0x00, 0x00, 0x00, 0x00, 0x16
};
unsigned int g_model_len = 19160;
```

複製此段程式（網路結構、權重及數量）到
tflite-micro-arduino-examples library

unsignd int g_model_len = 19160;

更新 Arduino 函式庫 (1/2)

This PC > Documents > Arduino > libraries > tflite-micro-arduino-examples-main > examples > micro_speech

Name	Date modified	Type	Size
data	2/23/2022 10:21 AM	File folder	
arduino_audio_provider	2/23/2022 10:21 AM	C++ Source File	7 KB
arduino_command_responder	2/23/2022 10:21 AM	C++ Source File	3 KB
arduino_main	2/23/2022 10:21 AM	C++ Source File	1 KB
audio_provider	2/23/2022 10:21 AM	C Header Source F...	3 KB
command_responder	2/23/2022 10:21 AM	C Header Source F...	2 KB
feature_provider	2/23/2022 10:21 AM	C++ Source File	6 KB
feature_provider	2/23/2022 10:21 AM	C Header Source F...	3 KB
main_functions	2/23/2022 10:21 AM	C Header Source F...	2 KB
micro_features_micro_features_generator	2/23/2022 10:21 AM	C++ Source File	5 KB
micro_features_micro_features_generator	2/23/2022 10:21 AM	C Header Source F...	2 KB
micro_features_micro_model_settings	2/23/2022 10:21 AM	C++ Source File	1 KB
micro_features_micro_model_settings	2/23/2022 10:21 AM	C Header Source F...	3 KB
micro_features_model	2/23/2022 10:21 AM	C++ Source File	118 KB
micro_features_model	2/23/2022 10:21 AM	C Header Source F...	2 KB
micro_speech	2/23/2022 10:21 AM	INO File	8 KB

- 找到函式庫原始路徑 Documents > Arduino > libraries > tflite-micro-arduino-examples
- 進到 examples > micro_speech
- 開啟 micro_features_model.cpp
- 將剛才訓練好的模型覆蓋掉。

```
const unsigned char g_model[] DATA_ALIGN_ATTRIBUTE = {
    0x20, 0x00, 0x00, 0x00, 0x54, 0x46, 0x4c, 0x33, 0x00, 0x00, 0x00, 0x00,
    0x00, 0x00, 0x12, 0x00, 0x1c, 0x00, 0x04, 0x00, 0x08, 0x00, 0x0c, 0x00,
    0x10, 0x00, 0x14, 0x00, 0x00, 0x00, 0x18, 0x00, 0x12, 0x00, 0x00, 0x00,
```

- 修改模型長度：
const int g_model_len = 19160;

更新 Arduino 函式庫 (2/2)

➤ **修改類別種類**：開啟 **micro_features_micro_model_settings.cpp**

```
#include "micro_features_micro_model_settings.h"
```

```
const char* kCategoryLabels[kCategoryCount] = {  
    "silence",  
    "unknown",  
    "yes",  
    "no",  
};
```

➤ **修改類別數量**：開啟 **micro_features_micro_model_settings.h**

```
constexpr int kCategoryCount = 4;
```

NO Code 測試結果

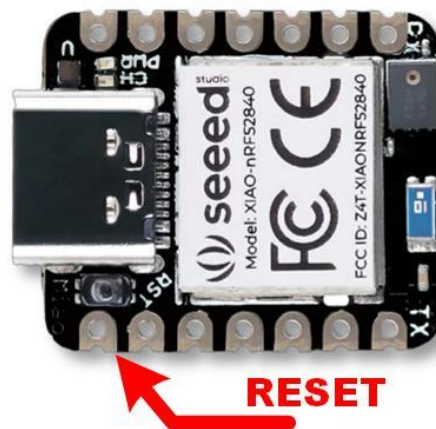
```
Initialization complete
Heard yes (205) @22400ms
Heard yes (212) @29920ms
Heard no (202) @84400ms
```

No Code

Virtual COM 輸出字串結果，不用改動任何一行程式碼。可令PC端或另一個MCU UART接收到結果後再自行解譯。

Low Code

如有必要可微調幾行驅動MCU上實體GPIO或其它週邊裝置。



RESET



OmniXRI整理製作, 2024/05/09

Arduino 2.0 快速(增量)編譯



如何讓Arduino 2.0 快速編譯 (增量編譯)

非常重要：

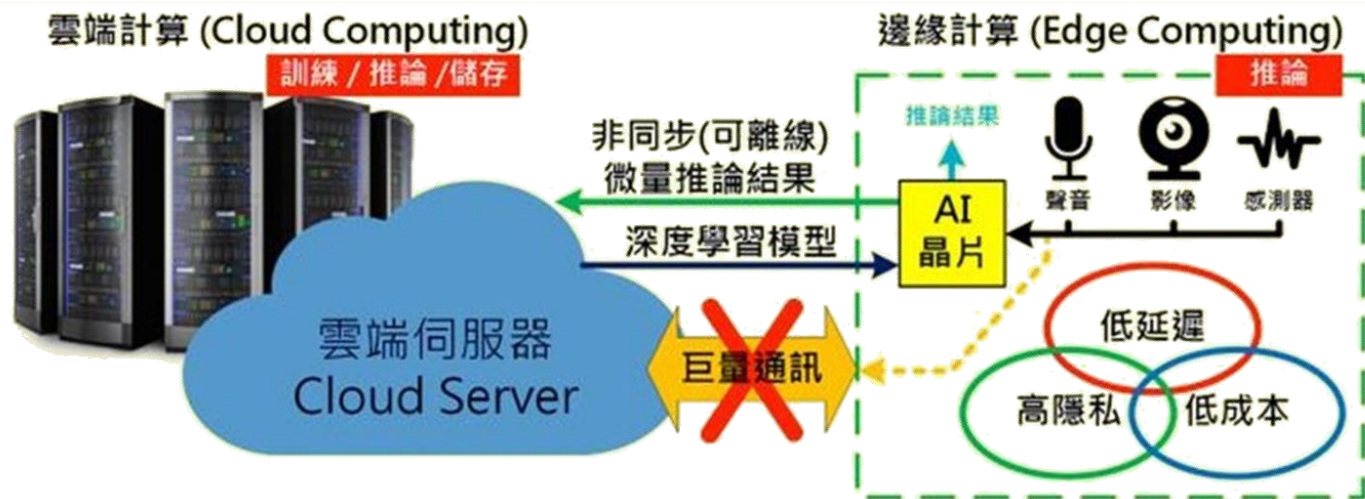
強烈建議一定要採用**Arduino CLI** 增量編譯方式更新程式修改，否則每次將耗費大量時間，增加碳排放量。

工作內容	Arduino IDE	Arduino CLI
第一次編譯上傳	19~20min	19~20min
未修改程式第二次編譯上傳	19~20min	< 1min
修改程式後編譯上傳	19~20min	1~2min

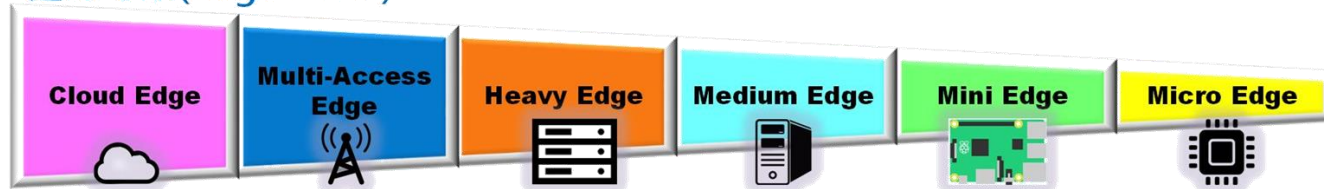
<https://omnixri.blogspot.com/2024/10/arduino-20.html>



Edge AI



邊緣等級(Edge Level)



8.3. 環境音辨識

ESC-50 環境音公開資料集

常見環境音，共分五大類（動物、自然聲及水聲、人聲及非講話聲、室內聲、戶外聲及都市雜音），每個分類又有十個小類，共五十分類。每個小類有四十個樣本，合計有兩千個樣本。

Animals	Natural soundscapes & water sounds	Human, non-speech sounds	Interior/domestic sounds	Exterior/urban noises
Dog	Rain	Crying baby	Door knock	Helicopter
Rooster	Sea waves	Sneezing	Mouse click	Chainsaw
Pig	Crackling fire	Clapping	Keyboard typing	Siren
Cow	Crickets	Breathing	Door, wood creaks	Car horn
Frog	Chirping birds	Coughing	Can opening	Engine
Cat	Water drops	Footsteps	Washing machine	Train
Hen	Wind	Laughing	Vacuum cleaner	Church bells
Insects (flying)	Pouring water	Brushing teeth	Clock alarm	Airplane
Sheep	Toilet flush	Snoring	Clock tick	Fireworks
Crow	Thunderstorm	Drinking, sipping	Glass breaking	Hand saw

ESC-50: Dataset for Environmental Sound Classification

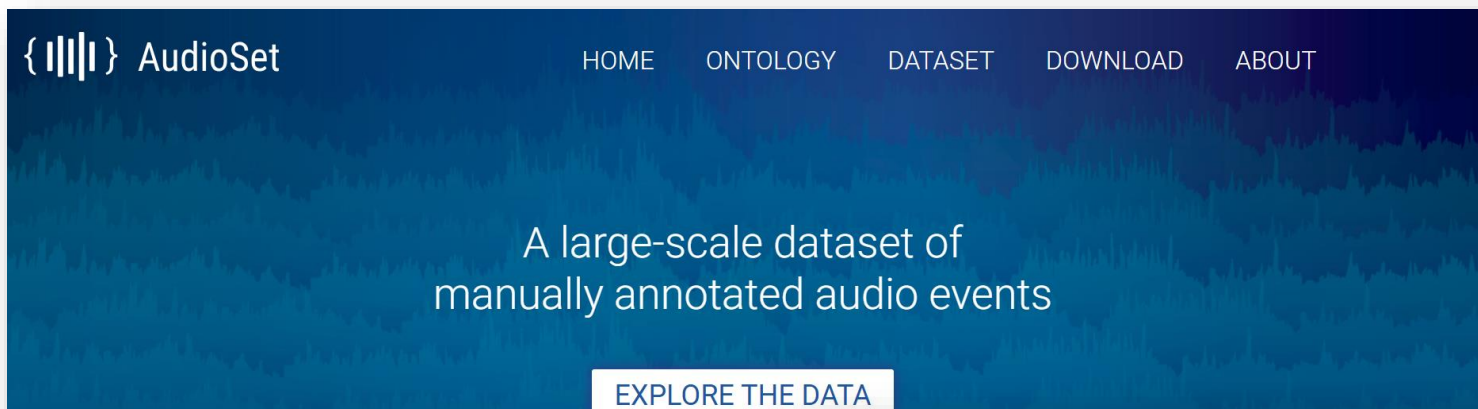
dog - 5-231762-A-0.wav



<https://github.com/karolpiczak/ESC-50>

Google AudioSet 聲音公開資料集

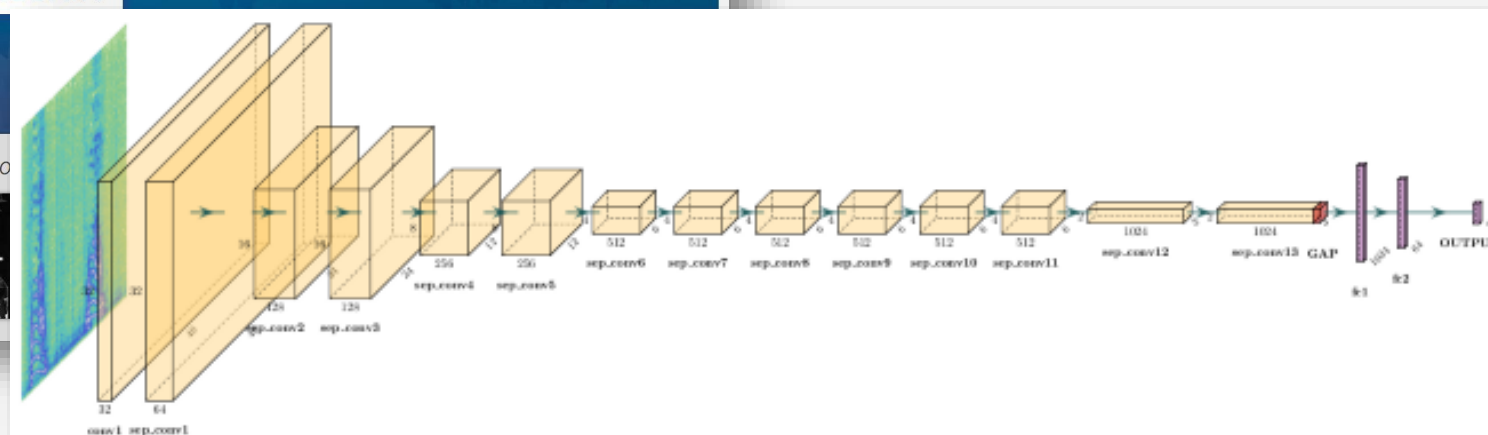
從Youtube中收集512類常見生活中聲音，可搭配 YAMNet 進行聲音分類。



Musical instrument (117,343 annotations in dataset)



Fireworks (3,051 annotations in dataset)



<https://research.google.com/audioset/>

<https://www.tensorflow.org/hub/tutorials/yamnet?hl=zh-cn>

案例分享 — 2024 總統盃黑客松卓越團隊



寧靜追蹤師Quiet Tracker

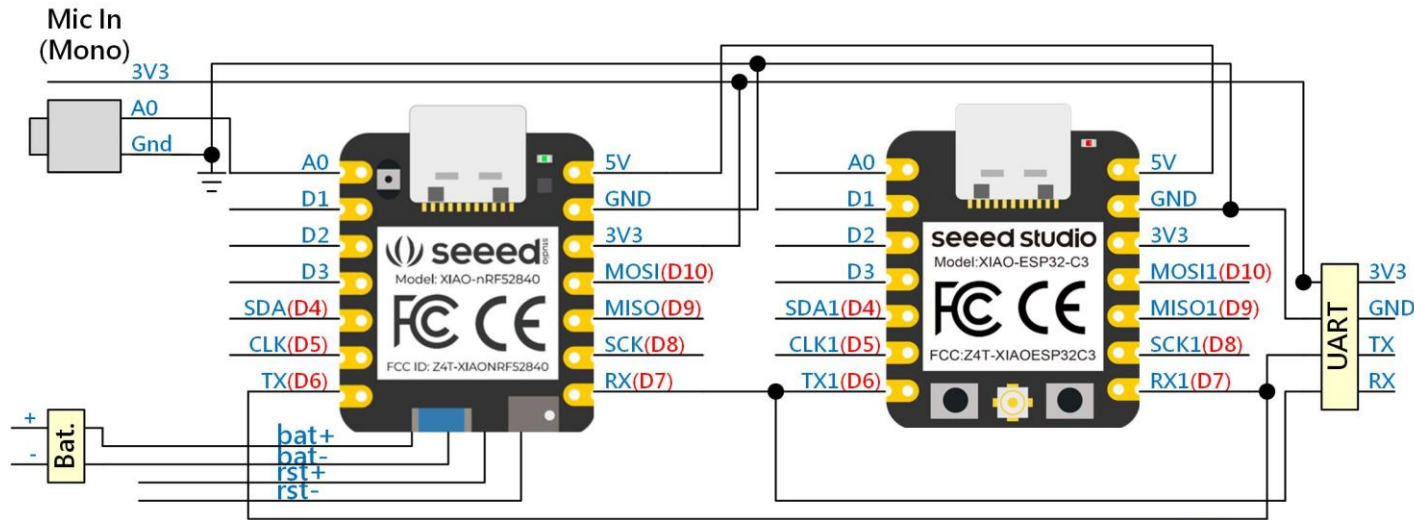
中央研究院人文社會科學研究中心詹大千研究員帶領下，共同打造結合大數據、低成本、可廣設之聲音盒子，藉由即時監測的聲音數據建構「安靜適宜性指數」(Quietness Suitability Index, QSI)地圖，未來可針對不易稽查、瞬間消逝的噪音進行即時監測並主動告發。



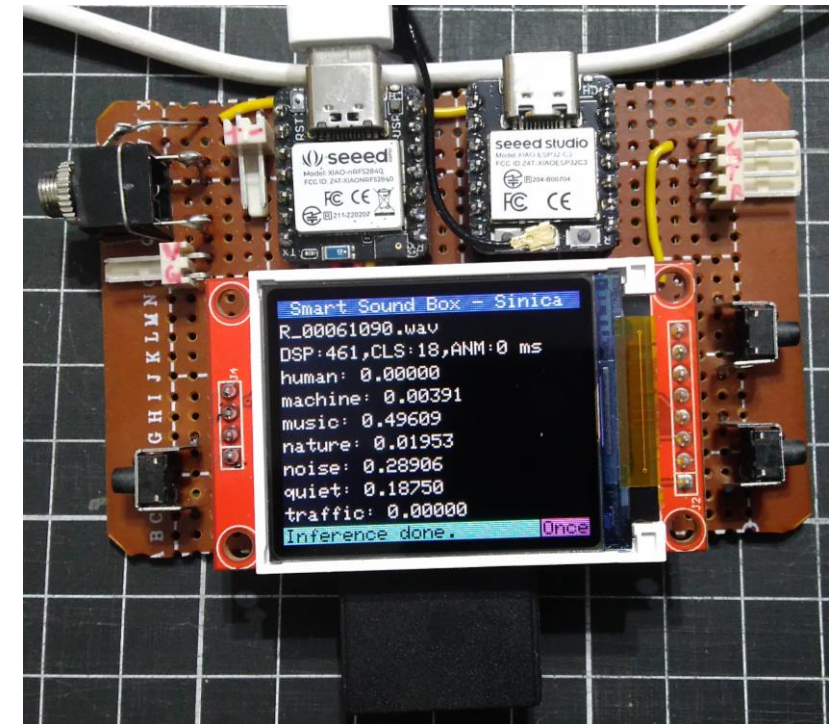
Youtub : <https://youtu.be/0E5gW9PcSgs>

資料來源 : <https://www.president.gov.tw/News/28977>

案例分享 – Smart Sound Box

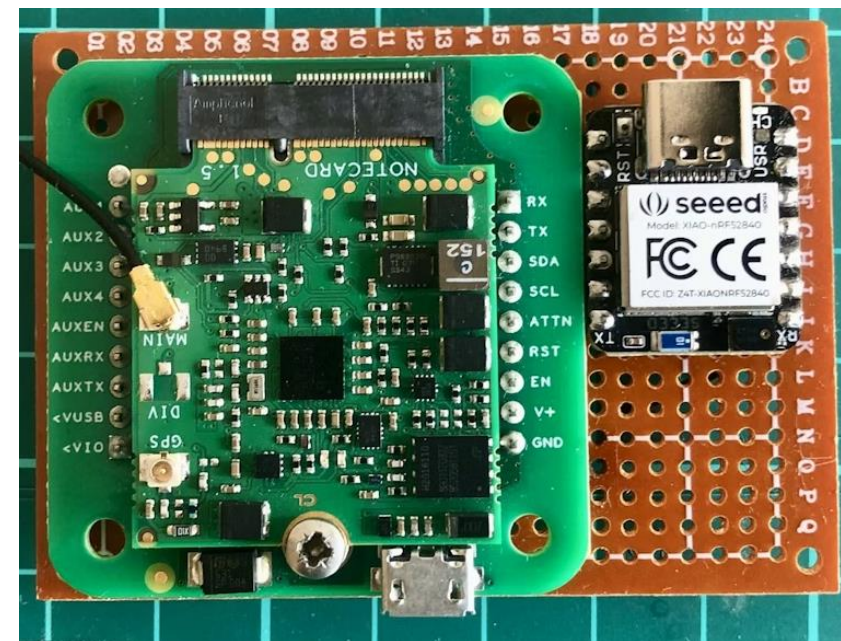
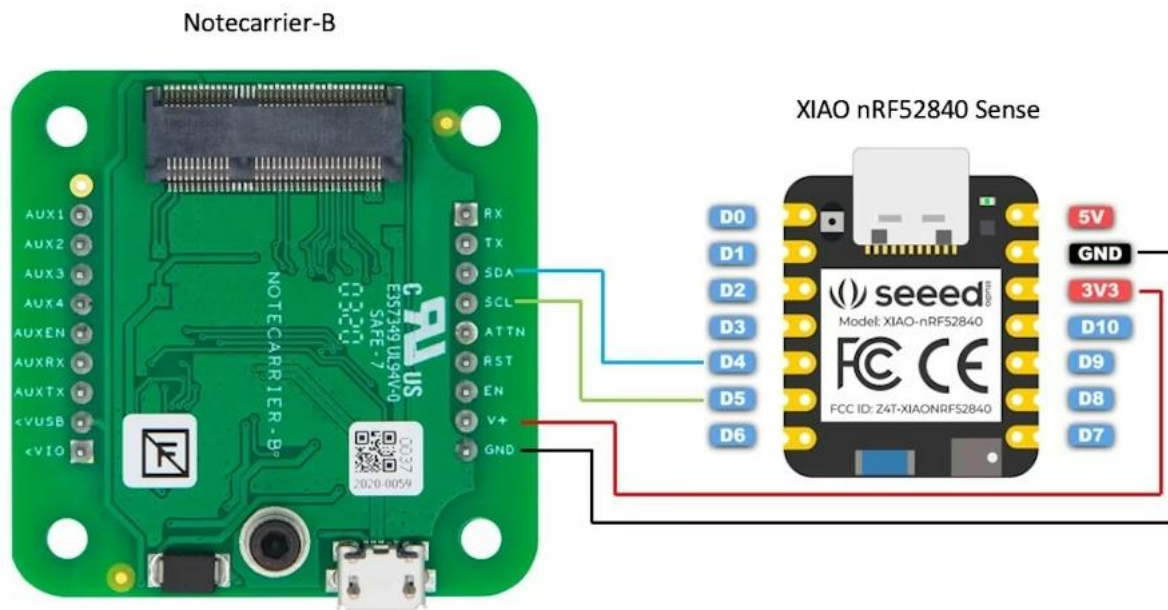


(間歇型聲音偵測)



案例分享—水流聲偵測

Running Faucet Detection – Seeed XIAO Sense + Blues Cellular (連續型聲音偵測)



Public Project Link:

<https://studio.edgeimpulse.com/public/119084/latest>

案例分享—更多聲音相關應用



1 Arduino BLE Sense 偵測
語音命令控制LED亮滅

2 CoreMaker-01 偵測語音
命令控制LED亮滅

3 Xiao nRF52840 Sense
關鍵詞偵測

4 寶寶哭聲自動搖籃

5 重金屬搖滾樂偵測器

6 智慧蓮蓬頭

7 智慧百葉窗

8 狗叫停止器



https://hackmd.io/@OmniXRI-Jack/tinyML_30_projects

參考文獻

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<https://omnixri.blogspot.com/p/ntust-edge-ai.html>
- 許哲豪，OmniXRI TinyML 小學堂 (2025) 【第 8 講】聲音辨識應用—環境音辨識
<https://www.youtube.com/watch?v=xXD8g-BXriU>
- 許哲豪，如何讓 Arduino 2.0 快速編譯 (增量編譯)
<https://omnixri.blogspot.com/2024/10/arduino-20.html>
- Seeed Xiao nRF52840 (Sense) 技術文件
https://wiki.seeedstudio.com/XIAO_BLE/
- 許哲豪，TinyML應用大全 (30組案例分享)
https://hackmd.io/@OmniXRI-Jack/tinyML_30_projects